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ELECTRIC AIRCRAFT

Abstract

In some embodiments, an electric aircraft may include a plurality of flight components including a plurality of control surfaces; a plurality of lift propulsors; at least a thrust propulsor; and a plurality of electric motors configured to power the plurality of propulsors; a flight controller, communicatively connected to a pilot input and flight components. In some embodiments, a flight controller may be configured to receive control datum from a pilot input; and generate an output datum as a function of the control datum.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority of U.S. Nonprovisional application Ser. No. 17/736,341, filed on May 4, 2022, and entitled “ELECTRIC AIRCRAFT,” and U.S. Nonprovisional application Ser. No. 17/736,357, filed on May 4, 2022, and entitled “ELECTRIC AIRCRAFT,” each of which is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

[0002] The present invention generally relates to the field of aircraft. In particular, the present invention is directed to electric aircraft.

BACKGROUND

[0003] In electrically propelled vehicles, such as an electric vertical takeoff and landing (eVTOL) aircraft, it is essential to maintain the integrity of the aircraft until safe landing. In some flights, a component of the aircraft may experience a malfunction or failure which will put the aircraft in an unsafe mode which will compromise the safety of the aircraft, passengers, and onboard cargo.

SUMMARY OF THE DISCLOSURE

[0004] In an aspect, an electric aircraft includes a fuselage having a fore end, an aft end, a first side, and a second side opposite the first side, and a longitudinal axis running from the fore end to the aft end and between the first side and the second side; a plurality of fixed wings, wherein the plurality of fixed wings includes a first wing, the first wing including a first proximal end affixed to the first side of the fuselage; a first distal end projecting away from the fuselage; a middle portion connecting the first proximal end to the first distal end; and a first aileron located on the first middle portion; and a second wing, the second wing including a second proximal end affixed to the second side of the fuselage; a second distal end projecting away from the fuselage; a second middle portion connecting the second proximal end to the second distal end; and a second aileron located on the second middle portion; a tail affixed to the aft end of the fuselage; a first boom including a first rear end affixed to the tail and a first forward end proximal to the fore end of the fuselage, wherein the first boom is affixed to the first wing; a second boom including a second rear end affixed to the tail and a second forward end proximal to the fore end of the fuselage, wherein the second boom is affixed to the second wing; and a plurality of lift components, wherein the plurality of lift components includes at least a first lift component mounted to the first boom; and at least a second lift component mounted to the second boom.

[0005] In another aspect, an electric aircraft includes a plurality of flight components wherein the plurality of flight components includes a plurality of control surfaces; a plurality of lift propulsors; at least a thrust propulsor; and a plurality of electric motors configured to power the plurality of propulsors; a flight controller, wherein flight controller is communicatively connected to a pilot input and flight components; configured to receive control datum from a pilot input; and generate an output datum as a function of the control datum.

[0006] The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0007] For the purpose of illustrating the invention, the drawings show aspects of one or more

embodiments of the invention. However, it should be understood that the present invention is not limited to the precise arrangements and instrumentalities shown in the drawings, wherein:

[0008] FIG. 1 is an aerial view of an exemplary electric aircraft;

[0009] FIG. 2 is an isometric view of an exemplary electric aircraft;

[0010] FIG. 3 is a block diagram of electronic communication of an electric aircraft;

[0011] FIG. 4 is a block diagram of an exemplary flight controller;

[0012] FIG. 5 is a diagrammatic representation of an exemplary embodiment of a reverse thrust command;

[0013] FIG. 6 is a diagrammatic representation of an exemplary embodiment of a flight mode;

[0014] FIG. 7 is an illustration of exemplary embodiments of net thrust angle of the propulsion system of an aircraft transitioning from fixed wing flight to vertical wing flight for landing;

[0015] FIG. 8 is a block diagram of an exemplary machine learning system; and

[0016] FIG. 9 is a block diagram of a computing system that can be used to implement any one or more of the methodologies disclosed herein and any one or more portions thereof.

[0017] The drawings are not necessarily to scale and may be illustrated by phantom lines, diagrammatic representations, and fragmentary views. In certain instances, details that are not necessary for an understanding of the embodiments or that render other details difficult to perceive may have been omitted. Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0018] At a high level, aspects of the present disclosure are directed to an electric aircraft. An electric aircraft may include a plurality of fixed wings, a tail, a first boom, and a second boom, and a plurality of lift components. A fuselage includes a fore end, an aft end, a first side, and a second side opposite the first side, along with a longitudinal axis running from the fore end to the aft end and between the first side and the second side. A plurality of fixed wings may include a first wing, the first wing including: a first proximal end affixed to the first side of the fuselage; a first distal end projecting away from the fuselage; a middle portion connecting the first proximal end to the first distal end; and a first aileron located on the first middle portion. A second wing may include a second proximal end affixed to the second side of the fuselage, a second distal end projecting away from the fuselage, a second middle portion connecting the second proximal end to the second distal end, and a second aileron located on the second middle portion. The aircraft also includes a tail affixed to the aft end of the fuselage. Additionally, a first boom may include a first rear end affixed to the tail and a first forward end proximal to the fore end of the fuselage, wherein the first boom is affixed to the first wing. A second boom may include a second rear end affixed to the tail and a second forward end proximal to the fore end of the fuselage, wherein the second boom is affixed to the second wing. The aircraft may include a plurality of lift components, wherein the plurality of lift components include at least a first lift component mounted to the first boom and at least a second lift component mounted to the second boom.

[0019] In an embodiment, aspects of the current disclosure are directed to electric aircraft comprised of a plurality of flight components and a flight controller. A plurality of flight components may be comprised of a plurality of control surfaces, a plurality of lift propulsors, at least a thrust propulsor, and a plurality of electric motors configured to power the plurality of propulsors. Flight controller may be communicatively connected to a pilot input and flight components. Flight controller may be configured to receive control datum from a pilot input and generate an output datum as a function of the control datum. Aspects of the present disclosure are directed to systems and methods for control of an aircraft's speed and altitude during fixed wing flight and/or rotor-based flight. Aspects of the present disclosure can also be used to describe the body of the electric aircraft. Exemplary embodiments illustrating aspects of the present disclosure are described below in the context of several specific examples.

[0020] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent,

however, that the present invention may be practiced without these specific details. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0021] Referring now to FIG. 1, an exemplary embodiment of an electric aircraft **100** which may be used in conjunction with system **100** of FIG. 1. For purposes of description herein, the terms “Proximal,” “Distal,” “Fore,” “Aft”, “Longitudinal Axis”, “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, “upward”, “downward”, “forward”, “backward” and derivatives thereof shall relate to the orientation in FIG. 1. The above mentioned terms are defined herein below. As used in the current disclosure, “Proximal” is defined as situated nearer to the fuselage or the point of attachment. “Distal” is defined as situated away from the fuselage or from the point of attachment. “Fore” is defined as situated or placed in front with regard to the direction of flight when in fixed-wing flight. “Aft” is defined at, near, or toward the tail of an aircraft. “Longitudinal axis” is an axis that passes through the aircraft from nose to tail. For the purposes of the current disclosure, components that are designated as the “first” are located on the right side of the aircraft as the aircraft is oriented in FIG. 1. Components designated as the “second” are on the left side of the aircraft as the aircraft is oriented in FIG. 1. When describing the “right” and “left” in the current disclosure it is assumed that the aircraft is oriented in the manner that it oriented in FIG. 1.

[0022] Still referring now to FIG. 1, the structure of the electric aircraft may include a fuselage **102**. As described herein, the fore end **102** of the fuselage is in the front of the aircraft near the cockpit. The aft end **104** of a fuselage is opposite of the fore end with the longitudinal axis **110** running through the center of the aft end **104**. The longitudinal axis runs through the middle of the fuselage parallel with the first boom **138** and second boom **140**. While facing the fore end **102** of the fuselage, the entire right side of the fuselage is the first side **106**, while the entire left side of the fuselage is the second side **108**.

[0023] Still referring now to FIG. 1, the structure of the electric aircraft may include a plurality of fixed wings. When the aircraft is oriented as depicted in FIG. 1, the wing on the right side of the aircraft is the first wing **112** and the wing on the left side is the second side **114**. The first wing **112** and second wing **114** both include a first wing proximal end **116** and a second wing proximal end **118**. The first and second proximal end **116/118** of the first and second wing **112/114** may be the portion of the wing nearest the fuselage as shown in FIG. 1. The first and second wing **112/114** both also include a first distal end **120** and second distal end **122**. The distal end **120/122** of the first and second wing **112/114** may be the portion of the wing further away from the fuselage. The first middle portion **124** and second middle portion **126** of the first and second wing **112/114** is located in the area between the proximal end **116/118** and the distal end **120/122** near the connection of the booms. The middle portion **124/126** of the first and second wing **112/114** may contain the first aileron **128** and second aileron **130**.

[0024] Still referring now to FIG. 1, the structure of the electric aircraft may include a tail **132**. Tail **132** may be mechanically attached to the aft end of the fuselage **104**. Tail **132** may include a first vertically projecting element **134** positioned to the first side of the fuselage **106**. Additionally tail **132** may include a second vertically projecting element **136** positioned to the second side of the

fuselage **108**.

[0025] Still referring now to FIG. **1**, the structure of the electric aircraft may include a first boom **138** and a second boom **140**. First and second boom **138/140** may be located near and/or affixed to the middle portion **126/124** of the first and second wing **114/112**. First and second boom **138/140** may be parallel with each other and the longitudinal axis **110**. First and second boom **138/140** may include a first rear end **142** and second rear end **144** affixed to the tail **132** near the aft end of the fuselage **104**. First and second boom **138/140** may also include a first forward end **146** and a second forward end **148**. The first and second boom forward end **146/148** may be located near to the fore end of the fuselage **102**. First boom **138** may be mechanically attached or affixed to the first wing **112**. Second boom **140** may also be mechanically attached or affixed to the second wing **114**.

[0026] Still referring now to FIG. **1**, the structure of the electric aircraft may include a plurality of lift components including first lift component **150** and second lift component **196**. A lift component may include a propulsor **212** as described in FIG. **2**. First lift component **150** are mounted on the first boom respectively **138**. Second lift component **196** may be mounted on the second boom **140**.

[0027] Referring now to FIG. **2**, an exemplary embodiment of an electric aircraft **200** is illustrated. Electric aircraft **200** may include an electrically powered aircraft. In some embodiments, electrically powered aircraft may be an electric vertical takeoff and landing (eVTOL) aircraft. Electric aircraft **200** may be capable of rotor-based cruising flight, rotor-based takeoff, rotor-based landing, fixed-wing cruising flight, airplane-style takeoff, airplane-style landing, and/or any combination thereof. "Rotor-based flight," as described in this disclosure, is where the aircraft generated lift and propulsion by way of one or more powered rotors coupled with an engine, such as a quadcopter, multi-rotor helicopter, or other vehicle that maintains its lift primarily using downward thrusting propulsors. "Fixed-wing flight," as described in this disclosure, is where the aircraft is capable of flight using wings and/or foils that generate lift caused by the aircraft's forward airspeed and the shape of the wings and/or foils, such as airplane-style flight.

[0028] Still referring to FIG. **2**, in some embodiments, fuselage **204** may comprise geodesic construction. Geodesic structural elements may include stringers wound about formers (which may be alternatively called station frames) in opposing spiral directions. A "stringer," as used herein, is a general structural element that comprises a long, thin, and rigid strip of metal or wood that is mechanically coupled to and spans the distance from, station frame to station frame to create an internal skeleton on which to mechanically couple aircraft skin. A former (or station frame) can include a rigid structural element that is disposed along the length of the interior of fuselage **204** orthogonal to the longitudinal (nose to tail) axis of the aircraft and forms the general shape of fuselage **204**. A former may comprise differing cross-sectional shapes at differing locations along fuselage **204**, as the former is the structural element that informs the overall shape of a fuselage **204** curvature. In embodiments, aircraft skin can be anchored to formers and strings such that the outer mold line of the volume encapsulated by the formers and stringers comprises the same shape as electric aircraft when installed. In other words, former(s) may form a fuselage's ribs, and the stringers may form the interstitials between such ribs. The spiral orientation of stringers about formers provides uniform robustness at any point on an aircraft fuselage such that if a portion sustains damage, another portion may remain largely unaffected. Aircraft skin would be mechanically coupled to underlying stringers and formers and may interact with a fluid, such as air, to generate lift and perform maneuvers.

[0029] Still referring to FIG. **2**, in an embodiment, fuselage **204** may comprise monocoque construction. Monocoque construction may include a primary structure that forms a shell (or skin in an aircraft's case) and supports physical loads. Monocoque fuselages are fuselages in which the aircraft skin or shell is also the primary structure. In monocoque construction aircraft skin would support tensile and compressive loads within itself and true monocoque aircraft can be further characterized by the absence of internal structural elements. Aircraft skin in this construction

method is rigid and can sustain its shape with no structural assistance from underlying skeleton-like elements. Monocoque fuselage may comprise aircraft skin made from plywood layered in varying grain directions, epoxy-impregnated fiberglass, carbon fiber, or any combination thereof.

[0030] Still referring to FIG. 2, in some embodiments, fuselage **204** may include a semi-monocoque construction. “Semi-monocoque construction,” as used herein, is a partial monocoque construction. In semi-monocoque construction, fuselage **204** may derive some structural support from stressed aircraft skin and some structural support from underlying frame structure made of structural elements. Formers or station frames can be seen running transverse to the long axis of fuselage **204** with circular cutouts which are generally used in real-world manufacturing for weight savings and for the routing of electrical harnesses and other modern on-board systems. In a semi-monocoque construction, stringers are the thin, long strips of material that run parallel to fuselage's long axis. Stringers may be mechanically coupled to formers permanently, such as with rivets. Aircraft skin may be mechanically coupled to stringers and formers permanently, such as by rivets as well. A person of ordinary skill in the art will appreciate that there are numerous methods for mechanical fastening of the aforementioned components like crews, nails, dowels, pins, anchors, adhesives like glue or epoxy, or bolts and nuts, to name a few. A subset of fuselage under the umbrella of semi-monocoque construction includes unibody vehicles. Unibody, which is short for “unitized body” or alternatively “unitary construction,” vehicles are characterized by a construction in which the body, floor plan, and chassis form a single structure. In the aircraft world, unibody may be characterized by internal structural elements like formers and stringers being constructed in one piece, integral to the aircraft skin as well as any floor construction like a deck.

[0031] Still referring to FIG. 2, stringers, and formers, which may account for the bulk of an aircraft structure excluding monocoque construction, may be arranged in a plurality of orientations depending on aircraft operation and materials. Stringers may be arranged to carry axial (tensile or compressive), shear, bending or torsion forces throughout their overall structure. Due to their coupling to aircraft skin, aerodynamic forces exerted on aircraft skin will be transferred to stringers. The location of said stringers greatly informs the type of forces and loads applied to each and every stringer, all of which may be handled by material selection, cross-sectional area, and mechanical coupling methods of each member. The same assessment may be made for formers. In general, formers are significantly larger in cross-sectional area and thickness, depending on location, than stringers. Both stringers and formers may comprise aluminum, aluminum alloys, graphite epoxy composite, steel alloys, titanium, or an undisclosed material alone or in combination.

[0032] Still referring to FIG. 2, in an embodiment, stressed skin, when used in semi-monocoque construction is the concept where the skin of an aircraft bears partial, yet significant, load in the overall structural hierarchy. In other words, the internal structure, whether it be a frame of welded tubes, formers and stringers, or some combination, is not sufficiently strong enough by design to bear all loads. The concept of stressed skin is applied in monocoque and semi-monocoque construction methods of fuselage **204**. Monocoque comprises only structural skin, and in that sense, aircraft skin undergoes stress by applied aerodynamic fluids imparted by the fluid. Stress as used in continuum mechanics can be described in pound-force per square inch (lbf/in²) or Pascals (Pa). In semi-monocoque construction stressed skin bears part of the aerodynamic loads and additionally may impart force on the underlying structure of stringers and formers.

[0033] Still referring to FIG. 2, it should be noted that an illustrative embodiment is presented only, and this disclosure in no way limits the form or construction of electric aircraft. In embodiments, fuselage **204** may be configurable based on the needs of the electric per specific mission or objective. The general arrangement of components, structural elements, and hardware associated with storing and/or moving a payload may be added or removed from fuselage **204** as needed, whether it is stowed manually, automatedly, or removed by personnel altogether. Fuselage **204** may be configurable for a plurality of storage options. Bulkheads and dividers may be installed and

uninstalled as needed, as well as longitudinal dividers where necessary. Bulkheads and dividers may be installed using integrated slots and hooks, tabs, boss and channel, or hardware like bolts, nuts, screws, nails, clips, pins, and/or dowels, to name a few. Fuselage **204** may also be configurable to accept certain specific cargo containers, or a receptacle that can, in turn, accept certain cargo containers.

[0034] Still referring to FIG. 2, electric aircraft may include a boom **208** or a plurality of booms **208** connected to fuselage **204**. As used in this disclosure a “boom” is an element that projects essentially horizontally from fuselage, including a laterally extending element, an outrigger, a spar, a lifting body, and/or a fixed wing **212** that extends from fuselage **204**. For the purposes of this disclosure, a “lifting body” is a structure that creates lift using aerodynamics. In some embodiments, boom **208** may be perpendicular to fuselage **204**. In some embodiments, a fixed wing may be mechanically attached to fuselage **204**. Fixed Wings **212** may be structures which include airfoils configured to create a pressure differential resulting in lift. Fixed Wings **212** may generally dispose on the left and right sides of the aircraft symmetrically, at a point between nose and empennage. Fixed Wings **212** may comprise a plurality of geometries in planform view, swept wing, tapered, variable wing, triangular, oblong, elliptical, square, among others. A wing's cross section may geometry comprises an airfoil. An “airfoil” as used in this disclosure is a shape specifically designed such that a fluid flowing above and below it exert differing levels of pressure against the top and bottom surface. In embodiments, the bottom surface of an aircraft can be configured to generate a greater pressure than does the top, resulting in lift. In an embodiment, and without limitation, fixed wing **212** may include a leading edge. As used in this disclosure a “leading edge” is a foremost edge of an airfoil that first intersects with the external medium. For example, and without limitation, leading edge may include one or more edges that may comprise one or more characteristics such as sweep, radius and/or stagnation point, droop, thermal effects, and the like thereof. In an embodiment, and without limitation, fixed wing **212** may include a trailing edge. As used in this disclosure a “trailing edge” is a rear edge of an airfoil. In an embodiment, and without limitation, trailing edge may include an edge capable of controlling the direction of the departing medium from the wing, such that a controlling force is exerted on the aircraft. Boom **208** may comprise differing and/or similar cross-sectional geometries over its cord length or the length from wing tip to where wing meets the aircraft's body. One or more fixed wing **212** may be symmetrical about the aircraft's longitudinal plane, which comprises the longitudinal or roll axis reaching down the center of the aircraft through the nose and empennage, and the plane's yaw axis. Boom may comprise control surfaces configured to be commanded by a pilot or pilots to change a fixed wing's **212** geometry and therefore its interaction with a fluid medium, like air. Control surfaces may comprise flaps, ailerons, tabs, spoilers, and slats, among others. The control surfaces may dispose on the wings in a plurality of locations and arrangements and in embodiments may be disposed at the leading and trailing edges of the wings, and may be configured to deflect up, down, forward, aft, or a combination thereof. An aircraft, including a dual-mode aircraft may comprise a combination of control surfaces to perform maneuvers while flying or on ground.

[0035] Still referring to FIG. 2, in an embodiment, a fixed wing **212** may include a plurality of control surfaces. As used in the current disclosure, “control surfaces” are aerodynamic devices attached to various points on an aircraft that allow a pilot to adjust and control the aircraft's flight attitude. Control surfaces may be configured to be commanded by a pilot or pilots to change a wing's geometry and therefore its interaction with a fluid medium, like air. In embodiments, control surfaces on a fixed-wing aircraft are attached to the airframe on hinges or tracks so they may move and thus deflect the air stream passing over them. This redirection of the air stream generates an unbalanced force to rotate the plane about the associated axis. There are three primary types of control surfaces an aileron, elevator/stabilator, and a rudder. Control surfaces may comprise flaps, ailerons, tabs, spoilers, and slats, among others. The control surfaces may dispose on the wings and tail in a plurality of locations and arrangements and in embodiments may be disposed at the leading

and trailing edges of the wings, and may be configured to deflect up, down, forward, aft, or a combination thereof. In other embodiments, control surfaces may be located on the tail of the aircraft primarily on the trailing edge.

[0036] Still referring to FIG. 2, in an embodiment, electric aircraft **200** may include a fuselage **204**. As used in this disclosure a “fuselage” is the main body of an aircraft, or in other words, the entirety of the aircraft except for the cockpit, nose, wings, boom, empennage, nacelles, any and all control surfaces, and generally contains an aircraft's payload. Fuselage **204** may comprise structural elements that physically support the shape and structure of an aircraft. Structural elements may take a plurality of forms, alone or in combination with other types. Structural elements may vary depending on the construction type of aircraft and specifically, the fuselage. Fuselage **204** may comprise a truss structure. A truss structure is often used with a lightweight aircraft and comprises welded steel tube trusses. A truss, as used herein, is an assembly of beams that create a rigid structure, often in combinations of triangles to create three-dimensional shapes. A truss structure may alternatively comprise wood construction in place of steel tubes, or a combination thereof. In embodiments, structural elements may comprise steel tubes and/or wood beams. In an embodiment, and without limitation, structural elements may include an aircraft skin. Aircraft skin may be layered over the body shape constructed by trusses. Aircraft skin may comprise a plurality of materials such as plywood sheets, aluminum, fiberglass, and/or carbon fiber, the latter of which will be addressed in greater detail later in this paper.

[0037] Still referring to FIG. 2, control surfaces may include an Aileron **208**. As used in the current disclosure, an “Aileron” is a hinged flight control surface usually forming part of the trailing edge of each wing of aircraft. Ailerons are used in pairs (one on each wing) to control the aircraft in roll (or movement around the aircraft's longitudinal axis), which normally results in a change in flight path due to the tilting of the lift vector. Whenever lift is increased, induced drag is also increased. An aileron may include any control surface mentioned in the current disclosure.

[0038] Still referring to FIG. 2, control surfaces may include an elevator. As used in the current disclosure, an “elevator” is a moveable part of the horizontal stabilizer, usually hinged to the back of the fixed part of the horizontal tail. Use of elevators control the plain around the pitch axis. The elevators move up and down together. In a non-limiting example, raised elevators push down on the tail and cause the nose to pitch up. This makes the wings fly at a higher angle of attack, which generates more lift and more drag.

[0039] Still referring to FIG. 2, control surfaces may include a rudder. As used in the current disclosure, a “rudder” is typically mounted on the trailing edge of the vertical stabilizer, part of the empennage. In a nonlimiting example, deflecting the rudder right pushes the tail left and causes the nose to yaw to the right. The reciprocal of the above mentioned example is also true. Centering the rudder pedals returns the rudder to neutral and stops the yaw.

[0040] Still referring to FIG. 2, electric aircraft may include a plurality of lift components **216** attached to boom **208**. As used in this disclosure a “lift component” is a component and/or device used to propel a craft upward by exerting downward force on a fluid medium, which may include a gaseous medium such as air or a liquid medium such as water. Lift component **216** may include any device or component that consumes electrical power on demand to propel an electric aircraft in a direction or other vehicle while on ground or in-flight. For example, and without limitation, lift component **216** may include a rotor, propeller, paddle wheel, and the like thereof, wherein a rotor is a component that produces torque along a longitudinal axis, and a propeller produces torque along a vertical axis. In an embodiment, lift component **216** may include a downward directed propulsor. In an embodiment, when a propulsor twists and pulls air behind it, it will, at the same time, push an aircraft forward with an equal amount of force. As a further non-limiting example, lift component **216** may include a thrust element which may be integrated into the propulsor. The thrust element may include, without limitation, a device using moving or rotating foils, such as one or more rotors, an airscrew or propeller, a set of airscrews or propellers such as contra-rotating propellers, a

moving or flapping wing, or the like. Further, a thrust element, for example, can include without limitation a marine propeller or screw, an impeller, a turbine, a pump-jet, a paddle or paddle-based device, or the like. The more air pulled behind an aircraft, the greater the force with which the aircraft is pushed forward. In some embodiments, the lift components **216** may include at least an electric motor, a propulsor assembly, and a motor nacelle. In some embodiments, the at least an electric motor and motor nacelle may be connected to boom **208**. As a non-limiting example, the at least an electric motor and motor nacelle may be indirectly connected to boom **208**.

[0041] Still referring to FIG. 2, in an embodiment, lift component **216** may include a plurality of blades. As used in this disclosure a “blade” is a propeller that converts rotary motion from an engine or other power source into a swirling slipstream. In an embodiment, blade may convert rotary motion to push the propeller forwards or backwards. In an embodiment lift component **216** may include a rotating power-driven hub, to which are attached several radial airfoil-section blades such that the whole assembly rotates about a longitudinal axis. The blades may be configured at an angle of attack. In an embodiment, and without limitation, angle of attack may include a fixed angle of attack. As used in this disclosure an “fixed angle of attack” is fixed angle between the chord line of the blade and the relative wind. As used in this disclosure a “fixed angle” is an angle that is secured and/or unmovable from the attachment point. For example, and without limitation, fixed angle of attack may be 2.8° as a function of a pitch angle of 8.1° and a relative wind angle 5.4° . In another embodiment, and without limitation, angle of attack may include a variable angle of attack. As used in this disclosure a “variable angle of attack” is a variable and/or moveable angle between the chord line of the blade and the relative wind. As used in this disclosure a “variable angle” is an angle that is moveable from the attachment point. For example, and without limitation variable angle of attack may be a first angle of 4.7° as a function of a pitch angle of 7.1° and a relative wind angle 2.4° , wherein the angle adjusts and/or shifts to a second angle of 2.7° as a function of a pitch angle of 5.1° and a relative wind angle 2.4° . In an embodiment, angle of attack be configured to produce a fixed pitch angle. As used in this disclosure a “fixed pitch angle” is a fixed angle between a cord line of a blade and the rotational velocity direction. For example, and without limitation, fixed pitch angle may include 18° . In another embodiment fixed angle of attack may be manually variable to a few set positions to adjust one or more lifts of the aircraft prior to flight. In an embodiment, blades for an aircraft are designed to be fixed to their hub at an angle similar to the thread on a screw makes an angle to the shaft; this angle may be referred to as a pitch or pitch angle which will determine the speed of the forward movement as the blade rotates.

[0042] Still referring to FIG. 2, in an embodiment, lift component **216** may be configured to produce a lift. As used in this disclosure a “lift” is a perpendicular force to the oncoming flow direction of fluid surrounding the surface. For example, and without limitation relative air speed may be horizontal to electric aircraft, wherein the lift force may be a force exerted in the vertical direction, directing electric aircraft upwards. In an embodiment, and without limitation, lift component **216** may produce lift as a function of applying a torque to lift component. As used in this disclosure a “torque” is a measure of force that causes an object to rotate about an axis in a direction. For example, and without limitation, torque may rotate an aileron and/or rudder to generate a force that may adjust and/or affect altitude, airspeed velocity, groundspeed velocity, direction during flight, and/or thrust. In an embodiment, and without limitation, lift component **216** may receive a source of power and/or energy from a power source may apply a torque on lift component **216** to produce lift. As used in this disclosure a “power source” is a source that that drives and/or controls any component attached to electric aircraft. For example, and without limitation power source may include a motor that operates to move one or more lift components, to drive one or more blades, or the like thereof. A motor may be driven by direct current (DC) electric power and may include, without limitation, brushless DC electric motors, switched reluctance motors, induction motors, or any combination thereof. A motor may also include electronic speed controllers or other components for regulating motor speed, rotation direction, and/or dynamic

braking.

[0043] Still referring to FIG. 2, power source may include an energy source. An energy source may include, for example, a generator, a photovoltaic device, a fuel cell such as a hydrogen fuel cell, direct methanol fuel cell, and/or solid oxide fuel cell, an electric energy storage device (e.g. a capacitor, an inductor, and/or a battery). An energy source may also include a battery cell, or a plurality of battery cells connected in series into a module and each module connected in series or in parallel with other modules. Configuration of an energy source containing connected modules may be designed to meet an energy or power requirement and may be designed to fit within a designated footprint in an electric aircraft in which electric aircraft may be incorporated.

[0044] Still referring to FIG. 2, in an embodiment, an energy source may be used to provide a steady supply of electrical power to a load over the course of a flight by a vehicle or other electric aircraft. For example, the energy source may be capable of providing sufficient power for “cruising” and other relatively low-energy phases of flight. An energy source may also be capable of providing electrical power for some higher-power phases of flight as well, particularly when the energy source is at a high SOC, as may be the case for instance during takeoff. In an embodiment, the energy source may be capable of providing sufficient electrical power for auxiliary loads including without limitation, lighting, navigation, communications, de-icing, steering, or other systems requiring power or energy. Further, the energy source may be capable of providing sufficient power for controlled descent and landing protocols, including, without limitation, hovering, descent, or runway landing. As used herein the energy source may have high power density where the electrical power an energy source can usefully produce per unit of volume and/or mass is relatively high. The electrical power is defined as the rate of electrical energy per unit time. An energy source may include a device for which power that may be produced per unit of volume and/or mass has been optimized, at the expense of the maximal total specific energy density or power capacity, during design.

[0045] Still referring to FIG. 2, an energy source may include a plurality of energy sources, referred to herein as a module of energy sources. The module may include batteries connected in parallel or in series or a plurality of modules connected either in series or in parallel designed to deliver both the power and energy requirements of the application. Connecting batteries in series may increase the voltage of at least an energy source which may provide more power on demand.

[0046] Still referring to FIG. 2, according to some embodiments, an energy source may include an emergency power unit (EPU) (i.e., auxiliary power unit). As used in this disclosure an “emergency power unit” is an energy source as described herein that is configured to power an essential system for a critical function in an emergency, for instance without limitation when another energy source has failed, is depleted, or is otherwise unavailable. Exemplary non-limiting essential systems include navigation systems, such as MFD, GPS, VOR receiver or directional gyro, and other essential flight components, such as propulsors.

[0047] Still referring to FIG. 2, another exemplary flight component may include landing gear. Landing gear may be used for take-off and/or landing. Landing gear may be used to contact ground while aircraft is not in flight.

[0048] Still referring to FIG. 2, in an embodiment, aircraft **200** may include a pilot control **220**. As used in this disclosure, a “pilot control” is a mechanism or means which allows a pilot to monitor and control operation of aircraft such as its flight components (for example, and without limitation, pusher component, lift component and other components such as propulsion components). For example, and without limitation, pilot control **220** may include a collective, inceptor, foot brake, steering and/or control wheel, control stick, pedals, throttle levers, and the like. Pilot control **220** may be configured to translate a pilot's desired torque for each flight component of the plurality of flight components, such as and without limitation, pusher component and lift component **216**. Pilot control **220** may be configured to control, via inputs and/or signals such as from a pilot, the pitch, roll, and yaw of the aircraft. Pilot control may be available onboard aircraft or remotely located

from it, as needed or desired.

[0049] Still referring to FIG. 2, as used in this disclosure a “collective control” or “collective” is a mechanical control of an aircraft that allows a pilot to adjust and/or control the pitch angle of plurality of flight components **208**. For example, and without limitation, collective control may alter and/or adjust the pitch angle of all of the main rotor blades collectively. For example, and without limitation pilot control **220** may include a yoke control. As used in this disclosure a “yoke control” is a mechanical control of an aircraft to control the pitch and/or roll. For example, and without limitation, yoke control may alter and/or adjust the roll angle of aircraft **200** as a function of controlling and/or maneuvering ailerons. In an embodiment, pilot control **220** may include one or more foot-brakes, control sticks, pedals, throttle levels, and the like thereof. In another embodiment, and without limitation, pilot control **220** may be configured to control a principal axis of the aircraft. As used in this disclosure a “principal axis” is an axis in a body representing one three dimensional orientations. For example, and without limitation, principal axis or more yaw, pitch, and/or roll axis. Principal axis may include a yaw axis. As used in this disclosure a “yaw axis” is an axis that is directed towards the bottom of aircraft, perpendicular to the wings. For example, and without limitation, a positive yawing motion may include adjusting and/or shifting nose of aircraft **200** to the right. Principal axis may include a pitch axis. As used in this disclosure a “pitch axis” is an axis that is directed towards the right laterally extending wing of aircraft. For example, and without limitation, a positive pitching motion may include adjusting and/or shifting nose of aircraft **200** upwards. Principal axis may include a roll axis. As used in this disclosure a “roll axis” is an axis that is directed longitudinally towards nose of aircraft, parallel to fuselage. For example, and without limitation, a positive rolling motion may include lifting the left and lowering the right wing concurrently. Pilot control **220** may be configured to modify a variable pitch angle. For example, and without limitation, pilot control **220** may adjust one or more angles of attack of a propulsor or propeller.

[0050] Still referring to FIG. 2, aircraft **200** may include at least an aircraft sensor **224**. Aircraft sensor **224** may include any sensor or noise monitoring circuit described in this disclosure. Aircraft sensor **224**, in some embodiments, may be communicatively connected or coupled to Flight controller. Aircraft sensor **224** may be configured to sense a characteristic of pilot control **220**. Sensor may be a device, module, and/or subsystem, utilizing any hardware, software, and/or any combination thereof to sense a characteristic and/or changes thereof, in an instant environment, for instance without limitation a pilot control **220**, which the sensor is proximal to or otherwise in a sensed communication with, and transmit information associated with the characteristic, for instance without limitation digitized data. Sensor **224** may be mechanically and/or communicatively coupled to aircraft **200**, including, for instance, to at least a pilot control **220**. Aircraft sensor **224** may be configured to sense a characteristic associated with at least a pilot control **220**. An environmental sensor may include without limitation one or more sensors used to detect ambient temperature, barometric pressure, and/or air velocity. Aircraft sensor **224** may include without limitation gyroscopes, accelerometers, inertial measurement unit (IMU), and/or magnetic sensors, one or more humidity sensors, one or more oxygen sensors, or the like. Additionally, or alternatively, sensor **224** may include at least a geospatial sensor. Aircraft sensor **224** may be located inside aircraft, and/or be included in and/or attached to at least a portion of aircraft. Sensor may include one or more proximity sensors, displacement sensors, vibration sensors, and the like thereof. Sensor may be used to monitor the status of aircraft **200** for both critical and non-critical functions. Sensor may be incorporated into vehicle or aircraft or be remote.

[0051] Still referring to FIG. 2, in some embodiments, aircraft sensor **224** may be configured to sense a characteristic associated with any pilot control described in this disclosure. Non-limiting examples of aircraft sensor **224** may include an inertial measurement unit (IMU), an accelerometer, a gyroscope, a proximity sensor, a pressure sensor, a light sensor, a pitot tube, an air speed sensor, a position sensor, a speed sensor, a switch, a thermometer, a strain gauge, an acoustic sensor, and an

electrical sensor. In some cases, aircraft sensor **224** may sense a characteristic as an analog measurement, for instance, yielding a continuously variable electrical potential indicative of the sensed characteristic. In these cases, aircraft sensor **224** may additionally comprise an analog to digital converter (ADC) as well as any additional circuitry, such as without limitation a Wheatstone bridge, an amplifier, a filter, and the like. For instance, in some cases, aircraft sensor **224** may comprise a strain gage configured to determine loading of one or more aircraft components, for instance landing gear. Strain gage may be included within a circuit comprising a Wheatstone bridge, an amplifier, and a bandpass filter to provide an analog strain measurement signal having a high signal to noise ratio, which characterizes strain on a landing gear member. An ADC may then digitize analog signal produces a digital signal that can then be transmitted other systems within aircraft **200**, for instance without limitation a computing system, a pilot display, and a memory component. Alternatively, or additionally, aircraft sensor **224** may sense a characteristic of a pilot control **220** digitally. For instance, in some embodiments, aircraft sensor **224** may sense a characteristic through a digital means or digitize a sensed signal natively. In some cases, for example, aircraft sensor **224** may include a rotational encoder and be configured to sense a rotational position of a pilot control; in this case, the rotational encoder digitally may sense rotational “clicks” by any known method, such as without limitation magnetically, optically, and the like. Aircraft sensor **224** may include any of the sensors as disclosed in the present disclosure. Aircraft sensor **224** may include a plurality of sensors. Any of these sensors may be located at any suitable position in or on aircraft **200**.

[0052] Still referring to FIG. 2, aircraft **200** may include at least a motor **228**, which may be mounted on a structural feature of the aircraft. Design of motor **228** may enable it to be installed external to structural member (such as a boom, nacelle, or fuselage) for easy maintenance access and to minimize accessibility requirements for the structure; this may improve structural efficiency by requiring fewer large holes in the mounting area. In some embodiments, motor **228** may include two main holes in top and bottom of mounting area to access bearing cartridge. Further, a structural feature may include a component of electric aircraft **200**. For example, and without limitation structural feature may be any portion of a vehicle incorporating motor **228**, including any vehicle as described in this disclosure. As a further non-limiting example, a structural feature may include without limitation a wing, a spar, an outrigger, a fuselage, or any portion thereof; persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of many possible features that may function as at least a structural feature. At least a structural feature may be constructed of any suitable material or combination of materials, including without limitation metal such as aluminum, titanium, steel, or the like, polymer materials or composites, fiberglass, carbon fiber, wood, or any other suitable material. As a non-limiting example, at least a structural feature may be constructed from additively manufactured polymer material with a carbon fiber exterior; aluminum parts or other elements may be enclosed for structural strength, or for purposes of supporting, for instance, vibration, torque, or shear stresses imposed by at least lift component. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various materials, combinations of materials, and/or constructions techniques.

[0053] Still referring to FIG. 2, aircraft **200** may include a motor nacelle. Motor nacelle surrounds the at least an electric motor. In an embodiment, motor nacelle may surround motor **228**. For the purposes of this disclosure, “motor nacelle” refers to a streamlined enclosure that houses an aircraft motor. In some embodiments, motor nacelle may be located on the wing or boom of an aircraft. In some other embodiments, motor nacelle may be part of an aircraft tail cone. A “tail cone,” for the purposes of this disclosure, refers to the conical section at the tail end of an aircraft.

[0054] Still referring to FIG. 2, electric aircraft **200** may include a vertical takeoff and landing aircraft (eVTOL). As used herein, a vertical take-off and landing (eVTOL) aircraft is one that can hover, take off, and land vertically. An eVTOL, as used herein, is an electrically powered aircraft typically using an energy source, of a plurality of energy sources to power the aircraft. In order to

optimize the power and energy necessary to propel the aircraft. eVTOL may be capable of rotor-based cruising flight, rotor-based takeoff, rotor-based landing, fixed-wing cruising flight, airplane-style takeoff, airplane-style landing, and/or any combination thereof. Rotor-based flight, as described herein, is where the aircraft generated lift and propulsion by way of one or more powered rotors coupled with an engine, such as a “quad copter,” multi-rotor helicopter, or other vehicle that maintains its lift primarily using downward thrusting propulsors. Fixed-wing flight, as described herein, is where the aircraft is capable of flight using wings and/or foils that generate lift caused by the aircraft's forward airspeed and the shape of the wings and/or foils, such as airplane-style flight. Boom may be located on aircraft **200**, attached and adjacent to the fuselage. Boom may extend perpendicularly to the fuselage.

[0055] Still referring to FIG. 2, a number of aerodynamic forces may act upon the electric aircraft during flight. Forces acting on electric aircraft **200** during flight may include, without limitation, thrust, the forward force produced by the rotating element of the electric aircraft and acts parallel to the longitudinal axis. Another force acting upon electric aircraft **200** may be, without limitation, drag, which may be defined as a rearward retarding force which is caused by disruption of airflow by any protruding surface of the electric aircraft **200** such as, without limitation, the wing, rotor, and fuselage. Drag may oppose thrust and acts rearward parallel to the relative wind. A further force acting upon electric aircraft **200** may include, without limitation, weight, which may include a combined load of the electric aircraft **200** itself, crew, baggage, and/or fuel. Weight may pull electric aircraft **200** downward due to the force of gravity. An additional force acting on electric aircraft **200** may include, without limitation, lift, which may act to oppose the downward force of weight and may be produced by the dynamic effect of air acting on the airfoil and/or downward thrust from the propulsor of the electric aircraft. Lift generated by the airfoil may depend on speed of airflow, density of air, total area of an airfoil and/or segment thereof, and/or an angle of attack between air and the airfoil. For example, and without limitation, electric aircraft **200** are designed to be as lightweight as possible. Reducing the weight of the aircraft and designing to reduce the number of components is essential to optimize the weight. To save energy, it may be useful to reduce weight of components of electric aircraft **200**, including without limitation propulsors and/or propulsion assemblies. In an embodiment, motor **228** may eliminate need for many external structural features that otherwise might be needed to join one component to another component. Motor **228** may also increase energy efficiency by enabling a lower physical propulsor profile, reducing drag and/or wind resistance. This may also increase durability by lessening the extent to which drag and/or wind resistance add to forces acting on electric aircraft **200** and/or propulsors.

[0056] Still referring to FIG. 2, electric aircraft may include at least a longitudinal thrust component. In some embodiments, longitudinal thrust component may be integrated with the tail cone of aircraft **200**. As used in this disclosure a “longitudinal thrust component” is a flight component that is mounted such that the component thrusts the flight component through a medium. As a non-limiting example, longitudinal thrust flight component may include a pusher flight component such as a pusher propeller, a pusher motor, a pusher propulsor, and the like. Additionally, or alternatively, pusher flight component may include a plurality of pusher flight components. As a further non-limiting example, longitudinal thrust flight component may include a puller flight component such as a puller propeller, a puller motor, a puller propulsor, and the like. Additionally, or alternatively, puller flight component may include a plurality of puller flight components.

[0057] Still referring to FIG. 2, in some embodiments, electric aircraft **200** includes, or may be coupled to or communicatively connected to, flight controller **232**. As used in this disclosure a “flight controller” is a computing device of a plurality of computing devices dedicated to data storage, security, distribution of traffic for load balancing, and flight instruction. In embodiments, flight controller **232** may be installed in an aircraft, may control the aircraft remotely, and/or may include an element installed in the aircraft and a remote element in communication therewith.

Flight controller **232**, in an embodiment, is located within fuselage **204** of aircraft. In accordance with some embodiments, flight controller **232** is configured to operate a vertical lift flight (upwards or downwards, that is, takeoff or landing), a fixed wing flight (forward or backwards), a transition between a vertical lift flight and a fixed wing flight, and a combination of a vertical lift flight and a fixed wing flight. In an embodiment, flight controller **232** may be configured to control a plurality of lift components and thrust components. Flight controller **232** may be configured to turn off lift propulsor and turn on thrust propulsor for fixed wing based flight. In other embodiments, flight controller **232** may be configured to select between hover based landing and a gliding landing. [0058] Still referring to FIG. **2**, in an embodiment, and without limitation, Flight controller **232** may be configured to operate a fixed-wing flight capability. Flight controller **232** may operate the fixed-wing flight capability as a function of reducing applied torque on lift (propulsor) component **216**. In an embodiment, and without limitation, an amount of lift generation may be related to an amount of forward thrust generated to increase airspeed velocity, wherein the amount of lift generation may be directly proportional to the amount of forward thrust produced. Additionally, or alternatively, flight controller may include an inertia compensator. As used in this disclosure an “inertia compensator” is one or more computing devices, electrical components, logic circuits, processors, and the like there of that are configured to compensate for inertia in one or more lift (propulsor) components present in aircraft **100**.

[0059] In an embodiment, and still referring to FIG. **2**, Flight controller **232** may be configured to perform a reverse thrust command. As used in this disclosure a “reverse thrust command” is a command to perform a thrust that forces a medium towards the relative air opposing aircraft **100**. In another embodiment, flight controller **232** may be configured to perform a regenerative drag operation. As used in this disclosure a “regenerative drag operation” is an operating condition of an aircraft, wherein the aircraft has a negative thrust and/or is reducing in airspeed velocity. For example, and without limitation, regenerative drag operation may include a positive propeller speed and a negative propeller thrust. Regenerative drag operation may alternatively or additionally include any regenerative drag operation as described in U.S. Nonprovisional application Ser. No. 17/319,155.

[0060] In an embodiment, and still referring to FIG. **2**, Flight controller **232** may be configured to perform a corrective action as a function of a failure event. For example, and without limitation, a corrective action may include an action to reduce a yaw torque generated by a failure event.

[0061] Still referring to FIG. **2**, a failure datum may be generated as a function of the determination that one or more power sources is losing capacity to provide sufficient power to a downward directed propulsor such as a lift component; this may be determined based on any suitable measure of an energy source capacity and/or output. For instance, and without limitation, an output voltage of the energy source may reduce and/or collapse below a threshold level, a current output may reduce below a threshold level, and/or a detected internal resistance may increase unexpectedly. This may alternatively or additionally be detected by detection that one or more other downward directed propulsors are consuming less power and/or producing less thrust, torque, force, or the like, which may indicate that less power is being provided to one or more components. Sensor may be further configured to generate a failure datum of the flight component of an aircraft as a function of the failure event. Failure datum may include, as an example and without limitation, a determination that a propulsor is damaged or otherwise operating insufficiently, such as without limitation a determination that a propulsor such as a propeller is not generating torque, and/or that the propulsor and/or propeller is generating less torque than expected and/or necessary to produce a level of thrust required to maintain airspeed and/or lift. As a further example a degree of torque may be sensed, without limitation, using load sensors deployed at and/or around a propulsor and/or by measuring back electromotive force (back EMF) generated by a motor driving the propulsor.

[0062] In an embodiment, and still referring to FIG. **2**, flight controller may be configured to receive failure datum from a sensor associated with a downward directed propulsor such as a lift

propulsor and determine a corrective action for a flight component of the plurality of flight components as a function of the failure datum. As used in this disclosure a “corrective action” is an action conducted by the plurality of flight components to correct and/or alter a movement of an aircraft, wherein a flight component is a component that promotes flight and guidance of an aircraft as described below in detail. Corrective action may be determined as a function of reducing yaw torque generated by the downward directed propulsor. As a non-limiting example, a yaw torque directing the nose of an aircraft to the right of a vertical axis may be generated due to a rudder movement and/or shifting. In an embodiment, flight controller may be communicatively coupled to the plurality of flight components. For example, plurality of flight components may include a component used to affect the aircrafts' roll and pitch which may comprise one or more ailerons, defined herein as hinged surfaces which form part of the trailing edge of each wing in a fixed wing aircraft, and which may be moved via mechanical means such as without limitation servomotors, mechanical linkages, or the like, to name a few. As a further example, the plurality of flight components may include a rudder, which may include, without limitation, a segmented rudder. The rudder may function, without limitation, to control yaw of an aircraft. The rudder may allow the aircraft to change in the horizontal direction, without altering the vertical direction. In an embodiment the rudder may include a rudder travel limiter. As used in this disclosure a “rudder travel limiter” is a maximum limit the rudder may be deflected. For example, a rudder may be limited to an angle of no more than 30°. Additionally or alternatively, the plurality of flight components may include other flight control surfaces such as propulsors, rotating flight controls, or any other structural features which can adjust the movement of the aircraft. In an embodiment, plurality of flight components may be oriented at a flight angle. As used in this disclosure a “flight angle” is an angle of the flight components to allow for flight capabilities. For example, and without limitation flight angle may be 7° for a first rotor, wherein the flight angle for a second rotor may be 7°.

[0063] Still referring to FIG. 2, lift component **216** may produce torque along a longitudinal axis, or a propeller produces torque along a vertical axis. As used in this disclosure a “torque” is a measure of force that causes an object to rotate about an axis in a direction. For example, and without limitation, torque may rotate an aileron and/or rudder to generate a force that may adjust and/or affect altitude, airspeed velocity, groundspeed velocity, direction during flight, and/or thrust. In an embodiment, and without limitation, lift component **216** may receive a source of power and/or energy from a power source may apply a torque on lift component **216** to produce lift.

[0064] In an embodiment, and still referring to FIG. 2, system **200** may contain movement resisting elements. As used in the current disclosure, “movement resisting elements” may be any apparatus or procedure that is used to cancel the effect of torque and/or moment on the aircraft. In an embodiment, movement resisting elements cancel out the torque and/or moment created by lift elements as a function of movement datum. Examples of movement resisting elements may include offsetting the angle of the downward directed propulsors by a torque cancellation angle while alternating the rotation of the various propellers. This will be discussed in more detail below.

[0065] Still referring to FIG. 2, A nonlimiting example of a movement resisting element is placing rotational axis of the lift element/propulsors offset from a vertical axis. As used in this disclosure a “rotational axis” is circular movement of a propeller about a vertical axis. For example, a propeller may revolve around a shaft, wherein the shaft is oriented along the vertical axis. In an embodiment a propeller may convert rotary motion from an engine or other power source into a swirling slipstream which pushes the propeller forwards or backwards. Propulsor may include a rotating power-driven hub, to which are attached several radial airfoil-section blades such that the whole assembly rotates about a longitudinal axis. As a non-limiting example, the blade pitch of the propellers may be fixed, manually variable to a few set positions, automatically variable (e.g. a “constant-speed” type), and/or any combination thereof. In an embodiment, propellers for an aircraft are designed to be fixed to their hub at an angle similar to the thread on a screw makes an

angle to the shaft; this angle may be referred to as a pitch or pitch angle which will determine the speed of the forward movement as the blade rotates. Additionally, or alternatively, downward directed propulsor has a rotational axis offset from a vertical axis by torque-cancellation angle. As used in this disclosure a “torque-cancellation angle” is an angle at which one or more propulsors are oriented about the vertical axis to reduce and/or eliminate a yaw torque. As used in this disclosure a “yaw torque” is a torque exerted along the vertical axis of an aircraft, wherein the vertical axis has its origin at the center of gravity and is directed towards the bottom of the aircraft, perpendicular to the wings and to the fuselage reference line. As a non-limiting example, a yaw torque directing the nose of an aircraft to the right of the vertical axis may be generated due to a rudder movement and/or shifting.

[0066] In an embodiment, and still referring to FIG. 2, torque-cancellation angle may include a nominal angle. As used in this disclosure a “nominal angle” is an angle of the propulsor in a horizontal axis. For example, and without limitation, a nominal angle may include a 3° angle tilted forward and/or a 3° angle tilted backward. Additionally, or alternatively, torque-cancellation angle may include a canted angle. As used in this disclosure a “canted angle” is an angle of the propulsor in longitudinal direction. For example, and without limitation, a nominal angle may include a 5.5° angle tilted inward and/or a 5.5° angle tilted outward. In an embodiment, and without limitation, the plurality of downward directed propulsors **216** may be attached to fuselage **204** at a yaw-torque-cancellation angle that is a fixed angle. As used in this disclosure a “fixed angle” is an angle that is secured and/or unmovable from the attachment point. For example, and without limitation, a fixed angle may be an angle of 3.4° inward and/or 5.2° forward. As a further non-limiting example, a fixed angle may be an angle of 3 inward and/or 0.6° forward. In an embodiment the fixed angle may include the respective yaw-cancellation. For example, and without limitation, plurality of downward directed propulsor **216**. may include a first downward directed propulsor having a first yaw-torque-cancellation angle with respect to the vertical axis and a second downward directed propulsor having a second yaw-cancelation angle with respect to the vertical axis. Additionally, or alternatively, a first downward directed propulsor may be moveable to the yaw-torque-cancellation angle as a function of an actuator, wherein an actuator is described in detail below. For example, and without limitation a first downward directed propulsor may be angled at a first angle, wherein an actuator may rotate and/or shift the first downward directed propulsor to a yaw-torque-cancellation angle.

[0067] Still referring to FIG. 2, an exemplary embodiment of a system for a structure of an electric aircraft is illustrated. System includes a flight controller **232**. flight controller **232** may include any computing device as described in this disclosure, including without limitation a microcontroller, microprocessor, digital signal processor (DSP) and/or system on a chip (SoC) as described in this disclosure. Computing device may include, be included in, and/or communicate with a mobile device such as a mobile telephone or smartphone. flight controller **232** may include a single computing device operating independently, or may include two or more computing device operating in concert, in parallel, sequentially or the like; two or more computing devices may be included together in a single computing device or in two or more computing devices. flight controller **232** may interface or communicate with one or more additional devices as described below in further detail via a network interface device. Network interface device may be utilized for connecting flight controller **232** to one or more of a variety of networks, and one or more devices. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A

network may employ a wired and/or a wireless mode of communication. In general, any network topology may be used. Information (e.g., data, software etc.) may be communicated to and/or from a computer and/or a computing device. flight controller **232** may include but is not limited to, for example, a computing device or cluster of computing devices in a first location and a second computing device or cluster of computing devices in a second location. flight controller **232** may include one or more computing devices dedicated to data storage, security, distribution of traffic for load balancing, and the like. flight controller **232** may distribute one or more computing tasks as described below across a plurality of computing devices of computing device, which may operate in parallel, in series, redundantly, or in any other manner used for distribution of tasks or memory between computing devices. flight controller **232** may be implemented using a “shared nothing” architecture in which data is cached at the worker, in an embodiment, this may enable scalability of system **200** and/or computing device.

[0068] Still referring to FIG. 2, flight controller **232** may be designed and/or configured to perform any method, method step, or sequence of method steps in any embodiment described in this disclosure, in any order and with any degree of repetition. For instance, flight controller **232** may be configured to perform a single step or sequence repeatedly until a desired or commanded outcome is achieved; repetition of a step or a sequence of steps may be performed iteratively and/or recursively using outputs of previous repetitions as inputs to subsequent repetitions, aggregating inputs and/or outputs of repetitions to produce an aggregate result, reduction or decrement of one or more variables such as global variables, and/or division of a larger processing task into a set of iteratively addressed smaller processing tasks. flight controller **232** may perform any step or sequence of steps as described in this disclosure in parallel, such as simultaneously and/or substantially simultaneously performing a step two or more times using two or more parallel threads, processor cores, or the like; division of tasks between parallel threads and/or processes may be performed according to any protocol suitable for division of tasks between iterations. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various ways in which steps, sequences of steps, processing tasks, and/or data may be subdivided, shared, or otherwise dealt with using iteration, recursion, and/or parallel processing.

[0069] Still referring to FIG. 2, flight controller **232** may calculate control surface datum as a function of control datum. As used in the current disclosure, a “control surface datum” is a signal which directs the movement of the plurality of control surfaces. Control Surface datum may signal to control surfaces to move in coordinated fashion with each other. Control surface datum may be used to control any control surface on aircraft **100**. In an embodiment, control surface datum may be used to adjust the speed or altitude of an aircraft by adjusting control surfaces.

[0070] Still referring to FIG. 2, flight controller **232** may be configured to calculate control surface datum as a function of pilot input using a machine learning process. Machine-learning module may perform determinations, classification, and/or analysis steps, methods, processes, or the like as described in this disclosure using machine learning processes. A “machine learning process,” as used in this disclosure, is a process that automatically uses training data to generate an algorithm that will be performed by a computing device/module to produce a preflight battery temperature given data provided as inputs. As used in the current disclosure, “pilot input training data,” as used herein, is data containing correlations that a machine-learning process may use to model relationships between two or more categories of data. In some embodiments, the inputs into the machine learning process may include a signal that corresponds to the use of a collective, inceptor, foot brake, steering and/or control wheel, control stick, pedals, throttle levers, and the like. In a non-limiting example, training data may also include wind and weather considerations, data recorded previous flights, pilot inputs from previous flights, expert inputs, fly by wire systems, sensor data from previous flight. Training data may additionally include any pilot inputs. Training data may be used to train a machine learning model, which may be done by the flight controller and/or on another device and then transmitted to flight controller. In some embodiments, training data may be

generated via electronic communication between a flight controller and plurality of sensors. In other embodiments, training data may be communicated to a machine learning model from a remote device. Once the control surface machine learning process receives training data, it may be implemented in any manner suitable for generation of receipt, implementation, or generation of machine learning.

[0071] In an embodiment, and still referring to FIG. 2, aircraft **200** may include at least an actuator configured to move each propulsor of the plurality of downward directed propulsors **212**. As used in this disclosure an “actuator” is a motor that may adjust an angle and/or position of a the downward directed propulsors. For example, and without limitation, an actuator may adjust rotor 4° in the horizontal axis. As a further non limiting example, an actuator may adjust an a propulsor from a first vertically aligned angle to a yaw-torque-cancellation angle. For example, downward directed propulsor **212** may be attached to fuselage **204** at a first vertical axis, wherein the first vertical axis may include a 3° inward and/or 1.4° forward wherein an actuator motor may maneuver and/or shift the downward directed propulsor $\pm 15^\circ$ in the horizontal and/or longitudinal axis. In an embodiment, and without limitation, actuator may be commanded as a function of a flight controller **232**. Flight controller **232** may include and/or communicate with any computing device as described in this disclosure, including without limitation a microcontroller, microprocessor, digital signal processor (DSP) and/or system on a chip (SoC) as described in this disclosure. Further, flight controller **232** may include a single computing device operating independently, or may include two or more computing device operating in concert, in parallel, sequentially or the like; two or more computing devices may be included together in a single computing device or in two or more computing devices. In embodiments, flight controller **232** may be installed in an aircraft, may control the aircraft remotely, and/or may include an element installed in the aircraft and a remote element in communication therewith.

[0072] In an embodiment, and still referring to FIG. 2, flight controller **232** may include a reconfigurable hardware platform. A “reconfigurable hardware platform,” as used herein, is a component and/or unit of hardware that may be reprogrammed, such that, for instance, a data path between elements such as logic gates or other digital circuit elements may be modified to change an algorithm, state, logical sequence, or the like of the component and/or unit. This may be accomplished with such flexible high-speed computing fabrics as field-programmable gate arrays (FPGAs), which may include a grid of interconnected logic gates, connections between which may be severed and/or restored to program in modified logic. Reconfigurable hardware platform may be reconfigured to enact any algorithm and/or algorithm selection process received from another computing device and/or created using machine-learning and/or neural net processes as described below.

[0073] Still referring to FIG. 2, aircraft **200** may include a tail **236** that is mechanically connected to both the boom and the fuselage. As used in the current disclosure, a “tail” is a structure at the rear of an aircraft that provides stability during flight. In embodiments, the tail **236** may include an empennage which is a device that incorporates vertical and horizontal stabilizing surfaces which stabilize the flight dynamics of yaw and pitch, as well as housing control surfaces. Structurally, the empennage consists of the entire tail **236** assembly, including the tailfin, the tailplane, and the part of the fuselage to which these are attached. The front section of the tailplane is called the horizontal stabilizer and is used to provide pitch stability. In embodiments the horizontal stabilizer may contain a control surface such as a rudder. The rear section of the tailplane is called the elevator and is a movable airfoil that controls changes in pitch, the up-and-down motion of the aircraft's nose. In some aircraft the horizontal stabilizer and elevator are one unit, and to control pitch the entire unit moves as one. This is known as a stabilator or full-flying stabilizer. The vertical tail **236** structure has a fixed front section called the vertical stabilizer, used to control yaw, which is movement of the fuselage right to left motion of the nose of the aircraft. The rear section of the vertical fin is the rudder, a movable airfoil that is used to turn the aircraft's nose right or left. When used in

combination with the ailerons, the result is a banking turn, a coordinated turn, the essential feature of aircraft movement.

[0074] Still referring to FIG. 2, stabilizing surfaces of tail **236** may be configured to be positioned in a V-shaped formation on the aft end of the fuselage **104**. In embodiments, arrangement of the tail **236** control surfaces that replaces the traditional fin and horizontal surfaces with two surfaces set in a V-shaped configuration. In embodiments, the first vertically projecting element **134** of the V formation may be projecting from the first side of the fuselage **106**. The second vertically projecting element **136** of the V formation is projecting from on the second side fuselage **108**. The first side vertically projecting element **134** may be the right side of tail as the aircraft is oriented in FIG. 1. Whereas the second side vertically projecting element **136** may be the left side of tail as the aircraft is oriented in FIG. 1. The combination of these projections should make a V like shape. The structures in the V shaped formation may be positioned closer to the horizontal axis while still on either the first or the second side **106/108** of the fuselage **204**. The V-shaped structures could be positioned such that the first side **106** is laterally between it and the longitudinal axis **110**, where a given point is “laterally between” means if you draw a line parallel to the longitudinal axis that passes through that point, the line will pass between the object and the longitudinal axis. The entire fuselage **204** is laterally between if the point of the fuselage farthest from the longitudinal axis **110** is laterally between. The V-shaped configuration of tail **236** may additionally include vertical stabilizers above the V-shaped portion. Control surfaces may be located at any location on tail **236**. Rudders may be located on the vertical portion of the V shaped configuration. A first rudder **240** may be located on first vertically projecting element **134**. A second rudder **244** may be located on the second vertically projecting element **136**. First and second rudders **240/244** may be comprised of a plurality of rudders. The tail **236** may be made of the same materials as the fixed wings. In embodiments, the boom's first rear end **142** structure may be mechanically affixed to the tail **236** at the first side of the V-formation. The Boom's second rear end structure **144** may be mechanically affixed to the tail **236** at the second side of the V-formation.

[0075] Still referring to FIG. 2, as used in the current disclosure. a “propulsor” is a component and/or device used to propel a craft by exerting force on a fluid medium, which may include a gaseous medium such as air or a liquid medium such as water. In an embodiment, when a propulsor twists and pulls air behind it, it may, at the same time, push an aircraft forward with an amount of force and/or thrust. More air pulled behind an aircraft results in greater thrust with which the aircraft is pushed forward. Propulsor component may include any device or component that consumes electrical power on demand to propel an electric aircraft in a direction or other vehicle while on ground or in-flight. In an embodiment, propulsor component may include a puller component. As used in this disclosure a “Thrust propulsor” is a component that pushes and/or thrusts an aircraft through a medium. As a non-limiting example, thrust propulsor **248** may include a pusher propeller, a paddle wheel, a pusher motor, a pusher propulsor, and the like. Thrust propulsor **248** may be primarily used in fixed wing based flight. Thrust propulsor **248** may be located at the rear end of fuselage **204**. Additionally, or alternatively, thrust propulsor **248** may include a plurality of pusher flight components. Thrust propulsor **248** is configured to produce a forward thrust. As a non-limiting example, forward thrust may include a force to force aircraft to in a horizontal direction along the longitudinal axis. As a further non-limiting example, thrust propulsor **248** may twist and/or rotate to pull air behind it and, at the same time, push aircraft **200** forward with an equal amount of force. In an embodiment, and without limitation, the more air forced behind aircraft, the greater the thrust force with which the aircraft is pushed horizontally will be. In another embodiment, and without limitation, forward thrust may force aircraft **200** through the medium of relative air. Additionally, or alternatively, plurality of flight components may include one or more puller components. As used in this disclosure a “puller component” is a component that pulls and/or tows an aircraft through a medium. As a non-limiting example, puller component may include a flight component such as a puller propeller, a puller motor, a tractor propeller, a

puller propulsor, and the like. Additionally, or alternatively, puller component may include a plurality of puller flight components.

[0076] Still referring to FIG. 2, motor assembly includes propulsor. In embodiments, Propulsor may include an integrated rotor. As used herein, a rotor is a portion of an electric motor that rotates with respect to a stator of the electric motor, such as stator. Propulsor may include a device or component that consumes electrical power on demand to propel an aircraft or other vehicle while on ground and/or in flight. Propulsor may include one or more propulsive devices. In an embodiment, propulsor may include a thrust element which may be integrated into the propulsor. A thrust element may include any device or component that converts the mechanical energy of a motor, for instance in the form of rotational motion of a shaft, into thrust in a fluid medium. For example, a thrust element may include without limitation a marine propeller or screw, an impeller, a turbine, a pump-jet, a paddle or paddle-based device, or the like. As another non-limiting example, at least a propulsor may include an eight-bladed pusher propeller, such as an eight-bladed propeller mounted behind the engine to ensure the drive shaft is in compression. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various devices that may be used as at least a thrust element. As used herein, a propulsive device may include, without limitation, a device using moving or rotating foils, including without limitation one or more rotors, an airscrew or propeller, a set of airscrews or propellers such as contra-rotating propellers, a moving or flapping wing, or the like.

[0077] Still referring to FIG. 2, thrust propulsor **248** may include a thrust element which may be integrated into the propulsor. Thrust propulsor **248** may include, without limitation, a device using moving or rotating foils, such as one or more rotors, an airscrew or propeller, a set of airscrews or propellers such as contra-rotating propellers, a moving or flapping wing, or the like. Further, a Thrust propulsor **248**, for example, can include without limitation a marine propeller or screw, an impeller, a turbine, a pump-jet, a paddle or paddle-based device, or the like.

[0078] In another embodiment, and still referring to FIG. 2, propulsor may include a propeller, a blade, or any combination of the two. A propeller may function to convert rotary motion from an engine or other power source into a swirling slipstream which may push the propeller forwards or backwards. Propulsor may include a rotating power-driven hub, to which several radial airfoil-section blades may be attached, such that an entire whole assembly rotates about a longitudinal axis. As a non-limiting example, blade pitch of propellers may be fixed at a fixed angle, manually variable to a few set positions, automatically variable (e.g. a “constant-speed” type), and/or any combination thereof as described further in this disclosure. As used in this disclosure a “fixed angle” is an angle that is secured and/or substantially unmovable from an attachment point. For example, and without limitation, a fixed angle may be an angle of 2.2° inward and/or 1.7° forward. As a further non-limiting example, a fixed angle may be an angle of 3.6° outward and/or 2.7° backward. In an embodiment, propellers for an aircraft may be designed to be fixed to their hub at an angle similar to the thread on a screw makes an angle to the shaft; this angle may be referred to as a pitch or pitch angle which may determine a speed of forward movement as the blade rotates. Additionally or alternatively, propulsor component may be configured having a variable pitch angle. As used in this disclosure a “variable pitch angle” is an angle that may be moved and/or rotated. For example, and without limitation, propulsor component may be angled at a first angle of 3.3° inward, wherein propulsor component may be rotated and/or shifted to a second angle of 1.7° outward.

[0079] In an embodiment, and still referring to FIG. 2, lift component may be configured to produce a lift. As used in this disclosure a “lift” is a perpendicular force to the oncoming flow direction of fluid surrounding the surface. For example, and without limitation relative air speed may be horizontal to the aircraft, wherein lift force may be a force exerted in a vertical direction, directing the aircraft upwards. As used in this disclosure a “lift component” is a component that lifts an aircraft through a medium. In an embodiment, and without limitation, lift component may

produce lift as a function of applying a torque to lift component. As used in this disclosure a “torque” is a measure of force that causes an object to rotate about an axis in a direction. For example, and without limitation, torque may rotate an aileron and/or rudder to generate a force that may adjust and/or affect altitude, airspeed velocity, groundspeed velocity, direction during flight, and/or thrust. In some embodiments, lift component may be considered a puller component.

[0080] Still referring to FIG. 2, in some embodiments the tail **236** of the aircraft may contain a tail rotor on the aft end of the fuselage **204**. As used in this disclosure a “tail rotor” is a smaller rotor mounted vertically and/or near-vertically at aft end of the fuselage **204**. Tail rotor may rotate to generate a yaw thrust in the same direction as the main rotor's rotation. Tail rotor may be positioned at a distance from the aircraft's center of mass to allow for enough thrust and/or torque to rotate the aircraft in the yaw direction. Tail rotor may include an adjustable pitch. As used in this disclosure an “adjustable pitch” is a pitch of the tail rotor blades that may be varied to provide directional control of the tail rotor in the yaw axis. For example, and without limitation, the tail rotor may rotate an aircraft 3° in the positive direction of the yaw axis to maintain a flight path. In an embodiment, and without limitation, the tail rotor may be composed of a core made of aluminum honeycomb and/or plasticized paper honeycomb, covered in a skin made of aluminum, carbon fiber composite, and/or titanium. Tail rotor may be fixed and/or adjustable as a function of an actuator motor.

[0081] In an embodiment and still referring to FIG. 2, a plurality of lift components of plurality of flight components may be arranged in a quad copter orientation. As used in this disclosure a “quad copter orientation” is at least a lift component oriented in a geometric shape and/or pattern, wherein each of the lift components is located along a vertex of the geometric shape. For example, and without limitation, a square quad copter orientation may have four lift components oriented in the geometric shape of a square, wherein each of the four lift components are located along the four vertices of the square shape. As a further non-limiting example, a hexagonal quad copter orientation may have six lift components oriented in the geometric shape of a hexagon, wherein each of the six lift components are located along the six vertices of the hexagon shape. In an embodiment, and without limitation, quad copter orientation may include a first set of lift components and a second set of lift components, wherein the first set of lift components and the second set of lift components may include two lift components each, wherein the first set of lift components and a second set of lift components are distinct from one another. For example, and without limitation, the first set of lift components may include two lift components that rotate in a clockwise direction, wherein the second set of lift components may include two lift components that rotate in a counterclockwise direction. In an embodiment, and without limitation, the first set of lift components may be oriented along a line oriented 45° from the longitudinal axis of aircraft **100**. In another embodiment, and without limitation, the second set of lift components may be oriented along a line oriented 135° from the longitudinal axis, wherein the first set of lift components line and the second set of lift components are perpendicular to each other.

[0082] Still referring to FIG. 2, propellers of lift component may be configured to be parked in an aerodynamically efficient manner during fixed wing flight. As used in the current disclosure, the term “parked” refers to the propulsors being placed locked in a position parallel to boom as shown in FIG. 2. Lift propulsors may be used during flight modes that include hovering, vertical take-off and landing, and all rotor based flight. Lift components will be parked during all fixed wing based flight modes. In embodiments, a flight controller may signal to lift components that aircraft **200** is engaged in fixed wing flight. Once this signal is received by lift components the propulsors will be locked into the parked position. In other embodiments, lift components may be parked in any position that is aerodynamically efficient. As used in the current disclosure, “aerodynamically efficient” is a measure of a design's propensity to generate aerodynamic forces for efficient flight parameters. In some embodiments, the most relevant consideration of aerodynamically efficiency is the lift/drag ratio.

[0083] Still referring to FIG. 2, propellers of lift component may be parked or active based on the flight mode of the aircraft. In some embodiments, a method may include (1) determining an angular datum, (2) generating a trajectory command as a function of the angular datum, and (3) initiating a transition from hover to fixed wing flight and/or from fixed wing flight to hover or a parked position as a function of the trajectory command. As used herein, an “angular datum” is any data describing the position or movement of any component of electric aircraft, for example a propulsor. Angular datum may include information about a position or movement of a propulsor, for instance measured in relative rotations, degrees, radians, encoder units, degrees per second, or the like. Angular datum may include information about the speed of the aircraft during flight. In a non-limiting example, angular datum may include information regarding the angle at which the electric aircraft is approaching the ground. Angular datum may also include information about angle of attack, angular, rate, altitude, air velocity, movement of the aircraft, position of the aircraft, and the like. Trajectory commands may include pre-determined angular factors such as angular rate, velocity trajectories, air velocity, and the like. An exemplary transition from fixed wing flight to vertical wing flight for landing is described with reference to FIG. 7.

[0084] Still referring to FIG. 2, lift components and thrust propulsors may be separate flight components. In embodiments, lift components and thrust propulsors are two separate entities that separately perform the functions of lifting and thrusting aircraft **200** respectively. Separating these functions allow aircraft **200** to operate in a more efficient manner.

[0085] Still referring to FIG. 2, aircraft **200** may include a plurality of motor assembly and at least one boom to house said motor assembly. Motor assembly may include an electric, gas, etc. motor. Motor is driven by electric power wherein power have varying or reversing voltage levels. For example, motor may be driven by alternating current (AC) wherein power is produced by an alternating current generator or inverter. Lift components and/or thrust propulsors may be attached to a motor assembly. For the purposes of this disclosure, an “electric motor,” is a machine that converts electrical energy into mechanical energy. Each electric motor **124** in system **100** includes a stator and at least an inverter.

[0086] Still referring to FIG. 2, Motor assembly may include at least a stator. A “stator,” as used herein, is a stationary component of a motor and/or motor assembly. In an embodiment, stator includes at least a first magnetic element. As used herein, first magnetic element is an element that generates a magnetic field. For example, first magnetic element may include one or more magnets which may be assembled in rows along a structural casing component. Further, first magnetic element may include one or more magnets having magnetic poles oriented in at least a first direction. The magnets may include at least a permanent magnet. Permanent magnets may be composed of, but are not limited to, ceramic, alnico, samarium cobalt, neodymium iron boron materials, any rare earth magnets, and the like. Further, the magnets may include an electromagnet. As used herein, an electromagnet is an electrical component that generates magnetic field via induction; the electromagnet may include a coil of electrically conducting material, through which an electric current flow to generate the magnetic field, also called a field coil of field winding. A coil may be wound around a magnetic core, which may include without limitation an iron core or other magnetic material. The core may include a plurality of steel rings insulated from one another and then laminated together; the steel rings may include slots in which the conducting wire will wrap around to form a coil. A first magnetic element may act to produce or generate a magnetic field to cause other magnetic elements to rotate, as described in further detail below. Stator may include a frame to house components including at least a first magnetic element, as well as one or more other elements or components as described in further detail below. In an embodiment, a magnetic field can be generated by a first magnetic element and can comprise a variable magnetic field. In embodiments, a variable magnetic field may be achieved by use of an inverter, a controller, or the like. In an embodiment, stator comprises an inner and outer cylindrical surface; a plurality of magnetic poles may extend outward from the outer cylindrical surface of the stator. Inner

cylindrical surface and outer cylindrical surface are coaxial about an axis of rotation.

[0087] Still referring to FIG. 2, propulsor may include a second magnetic element, which may include one or more further magnetic elements. Second magnetic element generates a magnetic field designed to interact with first magnetic element. Second magnetic element may be designed with a material such that the magnetic poles of at least a second magnetic element are oriented in an opposite direction from first magnetic element. Affixed, as described herein, is the attachment, fastening, connection, and the like, of one component to another component. Second magnetic element may include any magnetic element suitable for use as a first magnetic element. For instance, and without limitation, second magnetic element may include a permanent magnet and/or an electromagnet. Second magnetic element may include magnetic poles oriented in a second direction opposite of the orientation of the poles of first magnetic element. In an embodiment, motor assembly incorporates stator with a first magnet element and second magnetic element. First magnetic element includes magnetic poles oriented in a first direction, a second magnetic element includes a plurality of magnetic poles oriented in the opposite direction than the plurality of magnetic poles in the first magnetic element.

[0088] Still referring to FIG. 2, Aircraft **200** may include a driveshaft that is mechanically affixed to propulsors. As used in the current disclosure, a “driveshaft” is a component for transmitting mechanical power, torque, and rotation. In an embodiment, a driveshaft may be configured to couple to motor that produces the power to propulsor that uses this mechanical power to rotate the propellers. This connection involves mechanically linking the two components. In a nonlimiting example, the driveshaft may be used to transfer torque between components that are separated by a distance, since different components must be in different locations in the aircraft. To allow for variations in the alignment and distance between the propulsor and the motor, driveshafts frequently incorporate one or more universal joints, jaw couplings, or rag joints, and sometimes a splined joint or prismatic joint.

[0089] Still referring to FIG. 2, in an embodiment, flight controller **232** may be configured to receive a control datum. As used in this disclosure, a “control datum” is any element that reflects a pilot input. As used in this disclosure, a “pilot input” is a mechanism or means which allows a pilot to monitor and control operation of aircraft such as its flight components (for example, and without limitation, pusher component, lift component, control surfaces, and other components such as propulsion components). For example, and without limitation, pilot input may include a collective, inceptor, foot brake, steering and/or control wheel, control stick, pedals, throttle levers, and the like. Pilot input may be configured to translate a pilot's desired torque for each flight component of the plurality of flight components, such as and without limitation, control surfaces, lift components **216**, and thrust propulsors **248**. Pilot input may be configured to control, via inputs and/or signals such as from a pilot, the pitch, roll, and yaw of the aircraft. Pilot input may be available onboard aircraft or remotely located from it, as needed or desired. As used in this disclosure, “remote” is a spatial separation between two or more elements, systems, components, or devices. Stated differently, two elements may be remote from one another if they are physically spaced apart. For example, and without limitation, Pilot input may transmit from a remote location a signal to aircraft **200** control operation of its flight components. Pilot input may be located on ground while the aircraft is in flight.

[0090] Referring now to FIG. 3, an exemplary block diagram depicting electronic communication for an electric aircraft. System **300** may include a plurality of battery modules **304**. A “battery module” contains plurality of battery cells that have been wired together in series, parallel, or a combination of series and parallel, wherein the “battery module” holds the battery cells in a fixed position.

[0091] Still referring to FIG. 3, battery module includes an electrochemical cell. For the purposes of this disclosure, an “electrochemical cell” is a device capable of generating electrical energy from chemical reactions or using electrical energy to cause chemical reactions. Further, voltaic or

galvanic cells are electrochemical cells that generate electric current from chemical reactions, while electrolytic cells generate chemical reactions via electrolysis. In some embodiments, battery module **304** may include cylindrical battery cells. For the purposes of this disclosure, cylindrical battery cells are round battery cells that have a larger height than diameter.

[0092] Still referring to FIG. **3**, battery module **304** may be electrically connected to inverter **308**. An “inverter,” for the purposes of this disclosure, is a frequency converter that converts DC power from battery **304** into AC power. Specifically, first inverter and/or second inverter may supply AC power to drive first electric motor **124** and/or second electric motor **124**. First inverter and/or second inverter may be entirely electronic or a combination of mechanical elements and electronic circuitry. First inverter and/or second inverter may allow for variable speed and torque of the motor based on the demands of the vehicle.

[0093] Still referring to FIG. **3**, inverter may be configured to supply AC power to a plurality of flight components **312**. Flight components may include but is not limited to thrust propulsors, lift propulsors, control surfaces, tail, ailerons, rudders, motors, and the like. Each of the plurality of flight components are described in greater detail herein.

[0094] Still referring to FIG. **3**, flight controller **316** may be used to control a plurality of flight components **312**. Flight controller **316** may send an analog or digital signal to the plurality of flight components **312** to aid in controlling the electric aircraft. Flight controller **316** may use a pilot input to send a signal to flight components **312**. Additionally, Flight controller **316** may be communicatively connected with both battery **304** and Inverter **308**. Flight controller **316** is described in greater detail herein.

[0095] Now referring to FIG. **4**, an exemplary embodiment **400** of a flight controller **404** is illustrated. Flight controller **404** may include and/or communicate with any computing device as described in this disclosure, including without limitation a microcontroller, microprocessor, digital signal processor (DSP) and/or system on a chip (SoC) as described in this disclosure. Further, flight controller **404** may include a single computing device operating independently, or may include two or more computing device operating in concert, in parallel, sequentially or the like; two or more computing devices may be included together in a single computing device or in two or more computing devices. In embodiments, flight controller **404** may be installed in an aircraft, may control the aircraft remotely, and/or may include an element installed in the aircraft and a remote element in communication therewith.

[0096] In an embodiment, and still referring to FIG. **4**, flight controller **404** may include a signal transformation component **408**. As used in this disclosure a “signal transformation component” is a component that transforms and/or converts a first signal to a second signal, wherein a signal may include one or more digital and/or analog signals. For example, and without limitation, signal transformation component **408** may be configured to perform one or more operations such as preprocessing, lexical analysis, parsing, semantic analysis, and the like thereof. In an embodiment, and without limitation, signal transformation component **408** may include one or more analog-to-digital convertors that transform a first signal of an analog signal to a second signal of a digital signal. For example, and without limitation, an analog-to-digital converter may convert an analog input signal to a 10-bit binary digital representation of that signal. In another embodiment, signal transformation component **408** may include transforming one or more low-level languages such as, but not limited to, machine languages and/or assembly languages. For example, and without limitation, signal transformation component **408** may include transforming a binary language signal to an assembly language signal. In an embodiment, and without limitation, signal transformation component **408** may include transforming one or more high-level languages and/or formal languages such as but not limited to alphabets, strings, and/or languages. For example, and without limitation, high-level languages may include one or more system languages, scripting languages, domain-specific languages, visual languages, esoteric languages, and the like thereof. As a further non-limiting example, high-level languages may include one or more algebraic

formula languages, business data languages, string and list languages, object-oriented languages, and the like thereof.

[0097] Still referring to FIG. 4, signal transformation component **408** may be configured to optimize an intermediate representation **412**. As used in this disclosure an “intermediate representation” is a data structure and/or code that represents the input signal. Signal transformation component **408** may optimize intermediate representation as a function of a data-flow analysis, dependence analysis, alias analysis, pointer analysis, escape analysis, and the like thereof. In an embodiment, and without limitation, signal transformation component **408** may optimize intermediate representation **412** as a function of one or more inline expansions, dead code eliminations, constant propagation, loop transformations, and/or automatic parallelization functions. In another embodiment, signal transformation component **408** may optimize intermediate representation as a function of a machine dependent optimization such as a peephole optimization, wherein a peephole optimization may rewrite short sequences of code into more efficient sequences of code. Signal transformation component **408** may optimize intermediate representation to generate an output language, wherein an “output language,” as used herein, is the native machine language of flight controller **404**. For example, and without limitation, native machine language may include one or more binary and/or numerical languages.

[0098] In an embodiment, and without limitation, signal transformation component **408** may include transform one or more inputs and outputs as a function of an error correction code. An error correction code, also known as error correcting code (ECC), is an encoding of a message or lot of data using redundant information, permitting recovery of corrupted data. An ECC may include a block code, in which information is encoded on fixed-size packets and/or blocks of data elements such as symbols of predetermined size, bits, or the like. Reed-Solomon coding, in which message symbols within a symbol set having q symbols are encoded as coefficients of a polynomial of degree less than or equal to a natural number k , over a finite field F with q elements; strings so encoded have a minimum hamming distance of $k+1$, and permit correction of $(q-k-1)/2$ erroneous symbols. Block code may alternatively or additionally be implemented using Golay coding, also known as binary Golay coding, Bose-Chaudhuri, Hocquenghem (BCH) coding, multidimensional parity-check coding, and/or Hamming codes. An ECC may alternatively or additionally be based on a convolutional code.

[0099] In an embodiment, and still referring to FIG. 4, flight controller **404** may include a reconfigurable hardware platform **416**. A “reconfigurable hardware platform,” as used herein, is a component and/or unit of hardware that may be reprogrammed, such that, for instance, a data path between elements such as logic gates or other digital circuit elements may be modified to change an algorithm, state, logical sequence, or the like of the component and/or unit. This may be accomplished with such flexible high-speed computing fabrics as field-programmable gate arrays (FPGAs), which may include a grid of interconnected logic gates, connections between which may be severed and/or restored to program in modified logic. Reconfigurable hardware platform **416** may be reconfigured to enact any algorithm and/or algorithm selection process received from another computing device and/or created using machine-learning processes.

[0100] Still referring to FIG. 4, reconfigurable hardware platform **416** may include a logic component **420**. As used in this disclosure a “logic component” is a component that executes instructions on output language. For example, and without limitation, logic component may perform basic arithmetic, logic, controlling, input/output operations, and the like thereof. Logic component **420** may include any suitable processor, such as without limitation a component incorporating logical circuitry for performing arithmetic and logical operations, such as an arithmetic and logic unit (ALU), which may be regulated with a state machine and directed by operational inputs from memory and/or sensors; logic component **420** may be organized according to Von Neumann and/or Harvard architecture as a non-limiting example. Logic component **420** may include, incorporate, and/or be incorporated in, without limitation, a microcontroller,

microprocessor, digital signal processor (DSP), Field Programmable Gate Array (FPGA), Complex Programmable Logic Device (CPLD), Graphical Processing Unit (GPU), general purpose GPU, Tensor Processing Unit (TPU), analog or mixed signal processor, Trusted Platform Module (TPM), a floating point unit (FPU), and/or system on a chip (SoC). In an embodiment, logic component **420** may include one or more integrated circuit microprocessors, which may contain one or more central processing units, central processors, and/or main processors, on a single metal-oxide-semiconductor chip. Logic component **420** may be configured to execute a sequence of stored instructions to be performed on the output language and/or intermediate representation **412**. Logic component **420** may be configured to fetch and/or retrieve the instruction from a memory cache, wherein a “memory cache,” as used in this disclosure, is a stored instruction set on flight controller **404**. Logic component **420** may be configured to decode the instruction retrieved from the memory cache to opcodes and/or operands. Logic component **420** may be configured to execute the instruction on intermediate representation **412** and/or output language. For example, and without limitation, logic component **420** may be configured to execute an addition operation on intermediate representation **412** and/or output language.

[0101] In an embodiment, and without limitation, logic component **420** may be configured to calculate a flight element **424**. As used in this disclosure a “flight element” is an element of datum denoting a relative status of aircraft. For example, and without limitation, flight element **424** may denote one or more torques, thrusts, airspeed velocities, forces, altitudes, groundspeed velocities, directions during flight, directions facing, forces, orientations, and the like thereof. For example, and without limitation, flight element **424** may denote that aircraft is cruising at an altitude and/or with a sufficient magnitude of forward thrust. As a further non-limiting example, flight status may denote that is building thrust and/or groundspeed velocity in preparation for a takeoff. As a further non-limiting example, flight element **424** may denote that aircraft is following a flight path accurately and/or sufficiently.

[0102] Still referring to FIG. **4**, flight controller **404** may include a chipset component **428**. As used in this disclosure a “chipset component” is a component that manages data flow. In an embodiment, and without limitation, chipset component **428** may include a northbridge data flow path, wherein the northbridge dataflow path may manage data flow from logic component **420** to a high-speed device and/or component, such as a RAM, graphics controller, and the like thereof. In another embodiment, and without limitation, chipset component **428** may include a southbridge data flow path, wherein the southbridge dataflow path may manage data flow from logic component **420** to lower-speed peripheral buses, such as a peripheral component interconnect (PCI), industry standard architecture (ICA), and the like thereof. In an embodiment, and without limitation, southbridge data flow path may include managing data flow between peripheral connections such as ethernet, USB, audio devices, and the like thereof. Additionally, or alternatively, chipset component **428** may manage data flow between logic component **420**, memory cache, and a flight component **422**. As used in this disclosure a “flight component” is a portion of an aircraft that can be moved or adjusted to affect one or more flight elements. For example, flight component **422** may include a component used to affect the aircrafts' roll and pitch which may comprise one or more ailerons. As a further example, flight component **432** may include a rudder to control yaw of an aircraft. In an embodiment, chipset component **428** may be configured to communicate with a plurality of flight components as a function of flight element **424**. For example, and without limitation, chipset component **428** may transmit to an aircraft rotor to reduce torque of a first lift propulsor and increase the forward thrust produced by a pusher component to perform a flight maneuver.

[0103] In an embodiment, and still referring to FIG. **4**, flight controller **204** may be configured to generate an autonomous function. As used in this disclosure an “autonomous function” is a mode and/or function of flight controller **404** that controls aircraft automatically. For example, and without limitation, autonomous function may perform one or more aircraft maneuvers, take offs, landings, altitude adjustments, flight leveling adjustments, turns, climbs, and/or descents. As a

further non-limiting example, autonomous function may adjust one or more airspeed velocities, thrusts, torques, and/or groundspeed velocities. As a further non-limiting example, autonomous function may perform one or more flight path corrections and/or flight path modifications as a function of flight element **424**. In an embodiment, autonomous function may include one or more modes of autonomy such as, but not limited to, autonomous mode, semi-autonomous mode, and/or non-autonomous mode. As used in this disclosure “autonomous mode” is a mode that automatically adjusts and/or controls aircraft and/or the maneuvers of aircraft in its entirety. For example, autonomous mode may denote that flight controller **404** will adjust the aircraft. As used in this disclosure a “semi-autonomous mode” is a mode that automatically adjusts and/or controls a portion and/or section of aircraft. For example, and without limitation, semi-autonomous mode may denote that a pilot will control the propulsors, wherein flight controller **404** will control the ailerons and/or rudders. As used in this disclosure “non-autonomous mode” is a mode that denotes a pilot will control aircraft and/or maneuvers of aircraft in its entirety.

[0104] In an embodiment, and still referring to FIG. **4**, flight controller **404** may generate autonomous function as a function of an autonomous machine-learning model. As used in this disclosure an “autonomous machine-learning model” is a machine-learning model to produce an autonomous function output given flight element **424** and a pilot signal **436** as inputs; this is in contrast to a non-machine learning software program where the commands to be executed are determined in advance by a user and written in a programming language. As used in this disclosure a “pilot signal” is an element of datum representing one or more functions a pilot is controlling and/or adjusting. For example, pilot signal **436** may denote that a pilot is controlling and/or maneuvering ailerons, wherein the pilot is not in control of the rudders and/or propulsors. In an embodiment, pilot signal **436** may include an implicit signal and/or an explicit signal. For example, and without limitation, pilot signal **436** may include an explicit signal, wherein the pilot explicitly states there is a lack of control and/or desire for autonomous function. As a further non-limiting example, pilot signal **436** may include an explicit signal directing flight controller **404** to control and/or maintain a portion of aircraft, a portion of the flight plan, the entire aircraft, and/or the entire flight plan. As a further non-limiting example, pilot signal **436** may include an implicit signal, wherein flight controller **404** detects a lack of control such as by a malfunction, torque alteration, flight path deviation, and the like thereof. In an embodiment, and without limitation, pilot signal **436** may include one or more explicit signals to reduce torque, and/or one or more implicit signals that torque may be reduced due to reduction of airspeed velocity. In an embodiment, and without limitation, pilot signal **436** may include one or more local and/or global signals. For example, and without limitation, pilot signal **436** may include a local signal that is transmitted by a pilot and/or crew member. As a further non-limiting example, pilot signal **436** may include a global signal that is transmitted by air traffic control and/or one or more remote users that are in communication with the pilot of aircraft. In an embodiment, pilot signal **436** may be received as a function of a tri-state bus and/or multiplexor that denotes an explicit pilot signal should be transmitted prior to any implicit or global pilot signal.

[0105] Still referring to FIG. **4**, autonomous machine-learning model may include one or more autonomous machine-learning processes such as supervised, unsupervised, or reinforcement machine-learning processes that flight controller **404** and/or a remote device may or may not use in the generation of autonomous function. As used in this disclosure “remote device” is an external device to flight controller **404**. Additionally, or alternatively, autonomous machine-learning model may include one or more autonomous machine-learning processes that a field-programmable gate array (FPGA) may or may not use in the generation of autonomous function. Autonomous machine-learning process may include, without limitation machine learning processes such as simple linear regression, multiple linear regression, polynomial regression, support vector regression, ridge regression, lasso regression, elasticnet regression, decision tree regression, random forest regression, logistic regression, logistic classification, K-nearest neighbors, support

vector machines, kernel support vector machines, naïve bayes, decision tree classification, random forest classification, K-means clustering, hierarchical clustering, dimensionality reduction, principal component analysis, linear discriminant analysis, kernel principal component analysis, Q-learning, State Action Reward State Action (SARSA), Deep-Q network, Markov decision processes, Deep Deterministic Policy Gradient (DDPG), or the like thereof.

[0106] In an embodiment, and still referring to FIG. 4, autonomous machine learning model may be trained as a function of autonomous training data, wherein autonomous training data may correlate a flight element, pilot signal, and/or simulation data to an autonomous function. For example, and without limitation, a flight element of an airspeed velocity, a pilot signal of limited and/or no control of propulsors, and a simulation data of required airspeed velocity to reach the destination may result in an autonomous function that includes a semi-autonomous mode to increase thrust of the propulsors. Autonomous training data may be received as a function of user-entered valuations of flight elements, pilot signals, simulation data, and/or autonomous functions. Flight controller **404** may receive autonomous training data by receiving correlations of flight element, pilot signal, and/or simulation data to an autonomous function that were previously received and/or determined during a previous iteration of generation of autonomous function. Autonomous training data may be received by one or more remote devices and/or FPGAs that at least correlate a flight element, pilot signal, and/or simulation data to an autonomous function. Autonomous training data may be received in the form of one or more user-entered correlations of a flight element, pilot signal, and/or simulation data to an autonomous function.

[0107] Still referring to FIG. 4, flight controller **404** may receive autonomous machine-learning model from a remote device and/or FPGA that utilizes one or more autonomous machine learning processes, wherein a remote device and an FPGA is described above in detail. For example, and without limitation, a remote device may include a computing device, external device, processor, FPGA, microprocessor, and the like thereof. Remote device and/or FPGA may perform the autonomous machine-learning process using autonomous training data to generate autonomous function and transmit the output to flight controller **404**. Remote device and/or FPGA may transmit a signal, bit, datum, or parameter to flight controller **404** that at least relates to autonomous function. Additionally, or alternatively, the remote device and/or FPGA may provide an updated machine-learning model. For example, and without limitation, an updated machine-learning model may be comprised of a firmware update, a software update, an autonomous machine-learning process correction, and the like thereof. As a non-limiting example, a software update may incorporate a new simulation data that relates to a modified flight element. Additionally, or alternatively, the updated machine learning model may be transmitted to the remote device and/or FPGA, wherein the remote device and/or FPGA may replace the autonomous machine-learning model with the updated machine-learning model and generate the autonomous function as a function of the flight element, pilot signal, and/or simulation data using the updated machine-learning model. The updated machine-learning model may be transmitted by the remote device and/or FPGA and received by flight controller **404** as a software update, firmware update, or corrected autonomous machine-learning model. For example, and without limitation, autonomous machine learning model may utilize a neural net machine-learning process, wherein the updated machine-learning model may incorporate a gradient boosting machine-learning process.

[0108] Still referring to FIG. 4, flight controller **404** may include, be included in, and/or communicate with a mobile device such as a mobile telephone or smartphone. Further, flight controller may communicate with one or more additional devices as described below in further detail via a network interface device. The network interface device may be utilized for commutatively connecting a flight controller to one or more of a variety of networks, and one or more devices. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the

Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. The network may include any network topology and can may employ a wired and/or a wireless mode of communication.

[0109] In an embodiment, and still referring to FIG. 4, flight controller **404** may include, but is not limited to, for example, a cluster of flight controllers in a first location and a second flight controller or cluster of flight controllers in a second location. Flight controller **404** may include one or more flight controllers dedicated to data storage, security, distribution of traffic for load balancing, and the like. Flight controller **404** may be configured to distribute one or more computing tasks as described below across a plurality of flight controllers, which may operate in parallel, in series, redundantly, or in any other manner used for distribution of tasks or memory between computing devices. For example, and without limitation, flight controller **404** may implement a control algorithm to distribute and/or command the plurality of flight controllers. As used in this disclosure a “control algorithm” is a finite sequence of well-defined computer implementable instructions that may determine the flight component of the plurality of flight components to be adjusted. For example, and without limitation, control algorithm may include one or more algorithms that reduce and/or prevent aviation asymmetry. As a further non-limiting example, control algorithms may include one or more models generated as a function of a software including, but not limited to Simulink by MathWorks, Natick, Massachusetts, USA. In an embodiment, and without limitation, control algorithm may be configured to generate an auto-code, wherein an “auto-code,” is used herein, is a code and/or algorithm that is generated as a function of the one or more models and/or software's. In another embodiment, control algorithm may be configured to produce a segmented control algorithm. As used in this disclosure a “segmented control algorithm” is control algorithm that has been separated and/or parsed into discrete sections. For example, and without limitation, segmented control algorithm may parse control algorithm into two or more segments, wherein each segment of control algorithm may be performed by one or more flight controllers operating on distinct flight components.

[0110] In an embodiment, and still referring to FIG. 4, control algorithm may be configured to determine a segmentation boundary as a function of segmented control algorithm. As used in this disclosure a “segmentation boundary” is a limit and/or delineation associated with the segments of the segmented control algorithm. For example, and without limitation, segmentation boundary may denote that a segment in the control algorithm has a first starting section and/or a first ending section. As a further non-limiting example, segmentation boundary may include one or more boundaries associated with an ability of flight component **432**. In an embodiment, control algorithm may be configured to create an optimized signal communication as a function of segmentation boundary. For example, and without limitation, optimized signal communication may include identifying the discrete timing required to transmit and/or receive the one or more segmentation boundaries. In an embodiment, and without limitation, creating optimized signal communication further includes separating a plurality of signal codes across the plurality of flight controllers. For example, and without limitation the plurality of flight controllers may include one or more formal networks, wherein formal networks transmit data along an authority chain and/or are limited to task-related communications. As a further non-limiting example, communication network may include informal networks, wherein informal networks transmit data in any direction. In an embodiment, and without limitation, the plurality of flight controllers may include a chain path, wherein a “chain path,” as used herein, is a linear communication path comprising a hierarchy that data may flow through. In an embodiment, and without limitation, the plurality of flight controllers may include an all-channel path, wherein an “all-channel path,” as used herein, is a communication path that is not restricted to a particular direction. For example, and without

limitation, data may be transmitted upward, downward, laterally, and the like thereof. In an embodiment, and without limitation, the plurality of flight controllers may include one or more neural networks that assign a weighted value to a transmitted datum. For example, and without limitation, a weighted value may be assigned as a function of one or more signals denoting that a flight component is malfunctioning and/or in a failure state.

[0111] Still referring to FIG. 4, the plurality of flight controllers may include a master bus controller. As used in this disclosure a “master bus controller” is one or more devices and/or components that are connected to a bus to initiate a direct memory access transaction, wherein a bus is one or more terminals in a bus architecture. Master bus controller may communicate using synchronous and/or asynchronous bus control protocols. In an embodiment, master bus controller may include flight controller 404. In another embodiment, master bus controller may include one or more universal asynchronous receiver-transmitters (UART). For example, and without limitation, master bus controller may include one or more bus architectures that allow a bus to initiate a direct memory access transaction from one or more buses in the bus architectures. As a further non-limiting example, master bus controller may include one or more peripheral devices and/or components to communicate with another peripheral device and/or component and/or the master bus controller. In an embodiment, master bus controller may be configured to perform bus arbitration. As used in this disclosure “bus arbitration” is method and/or scheme to prevent multiple buses from attempting to communicate with and/or connect to master bus controller. For example, and without limitation, bus arbitration may include one or more schemes such as a small computer interface system, wherein a small computer interface system is a set of standards for physical connecting and transferring data between peripheral devices and master bus controller by defining commands, protocols, electrical, optical, and/or logical interfaces. In an embodiment, master bus controller may receive intermediate representation 412 and/or output language from logic component 420, wherein output language may include one or more analog-to-digital conversions, low bit rate transmissions, message encryptions, digital signals, binary signals, logic signals, analog signals, and the like thereof described above in detail.

[0112] Still referring to FIG. 4, master bus controller may communicate with a slave bus. As used in this disclosure a “slave bus” is one or more peripheral devices and/or components that initiate a bus transfer. For example, and without limitation, slave bus may receive one or more controls and/or asymmetric communications from master bus controller, wherein slave bus transfers data stored to master bus controller. In an embodiment, and without limitation, slave bus may include one or more internal buses, such as but not limited to a/an internal data bus, memory bus, system bus, front-side bus, and the like thereof. In another embodiment, and without limitation, slave bus may include one or more external buses such as external flight controllers, external computers, remote devices, printers, aircraft computer systems, flight control systems, and the like thereof.

[0113] In an embodiment, and still referring to FIG. 4, control algorithm may optimize signal communication as a function of determining one or more discrete timings. For example, and without limitation master bus controller may synchronize timing of the segmented control algorithm by injecting high priority timing signals on a bus of the master bus control. As used in this disclosure a “high priority timing signal” is information denoting that the information is important. For example, and without limitation, high priority timing signal may denote that a section of control algorithm is of high priority and should be analyzed and/or transmitted prior to any other sections being analyzed and/or transmitted. In an embodiment, high priority timing signal may include one or more priority packets. As used in this disclosure a “priority packet” is a formatted unit of data that is communicated between the plurality of flight controllers. For example, and without limitation, priority packet may denote that a section of control algorithm should be used and/or is of greater priority than other sections.

[0114] Still referring to FIG. 4, flight controller 404 may also be implemented using a “shared nothing” architecture in which data is cached at the worker, in an embodiment, this may enable

scalability of aircraft and/or computing device. Flight controller **404** may include a distributor flight controller. As used in this disclosure a “distributor flight controller” is a component that adjusts and/or controls a plurality of flight components as a function of a plurality of flight controllers. For example, distributor flight controller may include a flight controller that communicates with a plurality of additional flight controllers and/or clusters of flight controllers. In an embodiment, distributed flight control may include one or more neural networks. For example, neural network also known as an artificial neural network, is a network of “nodes,” or data structures having one or more inputs, one or more outputs, and a function determining outputs based on inputs. Such nodes may be organized in a network, such as without limitation a convolutional neural network, including an input layer of nodes, one or more intermediate layers, and an output layer of nodes. Connections between nodes may be created via the process of “training” the network, in which elements from a training dataset are applied to the input nodes, a suitable training algorithm (such as Levenberg-Marquardt, conjugate gradient, simulated annealing, or other algorithms) is then used to adjust the connections and weights between nodes in adjacent layers of the neural network to produce the desired values at the output nodes. This process is sometimes referred to as deep learning.

[0115] Still referring to FIG. **4**, a node may include, without limitation, a plurality of inputs x_i that may receive numerical values from inputs to a neural network containing the node and/or from other nodes. Node may perform a weighted sum of inputs using weights w_i that are multiplied by respective inputs x_i . Additionally, or alternatively, a bias b may be added to the weighted sum of the inputs such that an offset is added to each unit in the neural network layer that is independent of the input to the layer. The weighted sum may then be input into a function q , which may generate one or more outputs y . Weight w_i applied to an input x_i may indicate whether the input is “excitatory,” indicating that it has strong influence on the one or more outputs y , for instance by the corresponding weight having a large numerical value, and/or a “inhibitory,” indicating it has a weak effect influence on the one more inputs y , for instance by the corresponding weight having a small numerical value. The values of weights w_i may be determined by training a neural network using training data, which may be performed using any suitable process as described above. In an embodiment, and without limitation, a neural network may receive semantic units as inputs and output vectors representing such semantic units according to weights w_i that are derived using machine-learning processes as described in this disclosure.

[0116] Still referring to FIG. **4**, flight controller may include a sub-controller **440**. As used in this disclosure a “sub-controller” is a controller and/or component that is part of a distributed controller as described above; for instance, flight controller **404** may be and/or include a distributed flight controller made up of one or more sub-controllers. For example, and without limitation, sub-controller **440** may include any controllers and/or components thereof that are similar to distributed flight controller and/or flight controller as described above. Sub-controller **440** may include any component of any flight controller as described above. Sub-controller **440** may be implemented in any manner suitable for implementation of a flight controller as described above. As a further non-limiting example, sub-controller **440** may include one or more processors, logic components and/or computing devices capable of receiving, processing, and/or transmitting data across the distributed flight controller as described above. As a further non-limiting example, sub-controller **440** may include a controller that receives a signal from a first flight controller and/or first distributed flight controller component and transmits the signal to a plurality of additional sub-controllers and/or flight components.

[0117] Still referring to FIG. **4**, flight controller may include a co-controller **444**. As used in this disclosure a “co-controller” is a controller and/or component that joins flight controller **404** as components and/or nodes of a distributor flight controller as described above. For example, and without limitation, co-controller **444** may include one or more controllers and/or components that are similar to flight controller **404**. As a further non-limiting example, co-controller **444** may

include any controller and/or component that joins flight controller **404** to distributor flight controller. As a further non-limiting example, co-controller **444** may include one or more processors, logic components and/or computing devices capable of receiving, processing, and/or transmitting data to and/or from flight controller **404** to distributed flight control system. Co-controller **444** may include any component of any flight controller as described above. Co-controller **444** may be implemented in any manner suitable for implementation of a flight controller as described above.

[0118] Still referring to FIG. **4**, in some embodiments, flight controller **404** may be designed and/or configured to perform any method, method step, or sequence of method steps in any embodiment described in this disclosure, in any order and with any degree of repetition. For instance, flight controller **204** may be configured to perform a single step or sequence repeatedly until a desired or commanded outcome is achieved; repetition of a step or a sequence of steps may be performed iteratively and/or recursively using outputs of previous repetitions as inputs to subsequent repetitions, aggregating inputs and/or outputs of repetitions to produce an aggregate result, reduction or decrement of one or more variables such as global variables, and/or division of a larger processing task into a set of iteratively addressed smaller processing tasks. Flight controller may perform any step or sequence of steps as described in this disclosure in parallel, such as simultaneously and/or substantially simultaneously performing a step two or more times using two or more parallel threads, processor cores, or the like; division of tasks between parallel threads and/or processes may be performed according to any protocol suitable for division of tasks between iterations. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various ways in which steps, sequences of steps, processing tasks, and/or data may be subdivided, shared, or otherwise dealt with using iteration, recursion, and/or parallel processing.

[0119] Now referring to FIG. **5**, an exemplary embodiment **500** of aircraft **100** performing a reverse thrust command is illustrated. Aircraft **100** may include an energy source **504**. An energy source **504** may include any of energy source **504** as described above, in reference to FIG. **1**. Energy source **504** may be configured to provide energy and/or power to a motor **508**. Motor **508** may include any of the motor **508** as described above, in reference to FIG. **1**. Motor **508** may be mechanically coupled and/or secured to a propulsor component **512**. Propulsor component **512** may include any of the propulsor component as described above, in reference to FIG. **1**. Motor **508** may command propulsor component **512** to operate in a forward speed mode **516**. As used in this disclosure a “forward speed mode” is a mode and/or setting denoting that a flight component is rotating and/or moving at a rate in a first direction. For example, and without limitation, forward speed mode may include a propeller that rotates around a shaft at 6,000 revolutions per minute in a clockwise direction. Forward speed mode **516** may generate a positive thrust **520**. As used in this disclosure a “forward thrust” is a thrust that forces an aircraft through a medium, such as a relative air surrounding aircraft **100**. For example, forward thrust may include a thrust of 111 N directed towards the rear of an aircraft to at least force aircraft to be propelled forward. As used in this disclosure a “relative air” is an external medium that surrounds aircraft **100**. Additionally or alternatively, motor **508** may command propulsor component **512** to operate in a reverse speed mode **528**. As used in this disclosure a “reverse speed mode” is a mode and/or setting denoting that a flight component is rotating and/or moving at a rate in an opposite direction to the first direction. For example, and without limitation, reverse speed mode may include a propeller that rotates around a shaft at 2,000 revolutions per minute in a counterclockwise direction. Reverse speed mode **528** may generate a reverse thrust **532**. As used in this disclosure a “reverse thrust” is a thrust that forces a medium towards the relative air opposing aircraft **100**. For example, reverse thrust **532** may include a thrust of 180 N directed towards the nose of aircraft **100** to at least repel and/or oppose relative air **524**.

[0120] Now referring to FIG. **6**, an exemplary embodiment **600** of a flight mode is illustrated. As used in this disclosure a “flight mode” is one or more operating modes aircraft **100** may perform. In

an embodiment and without limitation, flight mode may be determined as a function of a propeller thrust **604**. As used in this disclosure a “propeller thrust” is the force generated by a propeller. For example, propeller thrust may include a forward thrust and/or reverse thrust. Additionally, or alternatively, flight mode may include a propeller speed **608**. As used in this disclosure a “propeller speed” is the rate at which a propeller rotates about a longitudinal axis. For example, propeller speed may include a forward speed mode and/or reverse speed mode as described in detail above. In an embodiment and without limitation, flight mode may include a normal flight region **612**. As used in this disclosure a “normal flight region” is a region wherein propeller thrust and propeller speed are both positive values. Normal flight region **612** may include one or more climb operations **616**. As used in this disclosure a “climb operation” is an operating condition of aircraft **100**, wherein the aircraft is elevating in altitude. For example, and without limitation, climb operation **616** may include an operating condition wherein the propeller speed and propeller thrust are both elevated. Normal flight region **612** may include one or more cruise operations **620**. As used in this disclosure a “cruise operation” is an operating condition of aircraft **100**, wherein the aircraft is maintaining a steady altitude and/or propeller thrust **604**. For example, and without limitation, cruise operation **620** may include an operating condition wherein the propeller speed is elevated and the propeller thrust is stabilized to allow aircraft **100** to maintain a steady flight path without increasing and/or decreasing the aircraft speed.

[0121] In an embodiment, and still referring to FIG. **6**, flight mode may include a thrust reversal region **624**. As used in this disclosure a “thrust reversal region” is a region wherein propeller thrust is negative and/or reversed and propeller speed is positive. Thrust reversal region **624** may include one or more regenerative drag operations **628**. As used in this disclosure a “regenerative drag operation” is an operating condition of aircraft **100**, wherein the aircraft has a negative thrust and/or is reducing in airspeed velocity. For example, and without limitation, regenerative drag operation **628** may include a positive propeller speed and a negative propeller thrust. In an embodiment, and without limitation, flight mode may include a speed reversal region **632**. As used in this disclosure a “speed reversal region” is a region wherein propeller thrust and propeller speed is negative. Speed reversal region **632** may include one or more dissipative drag operations **636**. As used in this disclosure a “dissipative drag operation” is an operating condition of aircraft **100**, wherein the aircraft has a negative thrust and a negative propeller speed. In an embodiment, and without limitation, dissipative drag operation may include one or more operating conditions that result in a braking mechanism.

[0122] Now referring to FIG. **7**, an illustration **700** of exemplary embodiments of net thrust angle of the propulsion system of an aircraft transitioning from fixed wing flight to vertical wing flight for landing is provided. At net thrust angle **704** descent has not started. At net thrust angle **704**, the forward propulsors are producing an increased amount of torque in forward flight while the vertical propulsors are outputting a minimal amount of torque to maintain a constant altitude and a low angle of attack. In a non-limiting embodiment, the beginning stage of descent may be triggered based on a trajectory command from the flight controller. Flight controller may determine when to begin descent based on a target landing location. Flight controller may instruct the aircraft to maintain a dive or initial descent position for the majority of the landing approach. Prior to descent, the vertical propulsors of an aircraft may be in a neutral position. In the beginning stage of descent, the Flight controller may instruct vertical propulsors to decrease their torque output. As descent begins, the vertical propulsors begin to decrease its torque output to reduce the altitude and approach a landing location at net thrust angle **708**. In a non-limiting embodiment, a front pair of vertical propulsors that are coupled closer to the front of the aircraft may output significantly more torque than the pair of vertical propulsors to achieve a high angle of attack. In a non-limiting embodiment, forward propulsors may still output torque in a decreasing amount to provide some lift to the aircraft as it slowly descends to the landing location. The combination of wing and rotor lift with some forward speed may be able to support the vehicle down to a low landing speed.

Below the low landing speed, gravity may cause the aircraft to fall towards ground. The aircraft may apply a large amount of power to slow the aircraft as it falls, allowing the aircraft to slowly accelerate towards the ground in an arc. The trajectory of speed, control inputs, and altitude may be optimized to allow the aircraft to touch down accurately in a planned position. As initial descent begins, the vertical propulsors are increasing its torque output to reduce the descent speed of the aircraft at net thrust angle **712**. In a non-limiting embodiment, forward propulsor may be increasing its rate of decrease in torque output at net thrust angle **712**. The nose of the aircraft may be sharply angled away from ground. The aircraft may be in a full upwards tilt position, e.g. tilted as far as the aircraft controls or actuators allow. The trajectory of the aircraft may allow it to slow down before touching down. The surface of the wings may provide drag those aids in slowing the aircraft. The rotors vertical propulsors be powered at maximum power, pulling the aircraft backwards and away from ground. When a power surge is applied to slow down the aircraft, the vertical propulsors are each outputting the same amount of torque to maintain an upright position at net thrust angle **716**. In a non-limiting embodiment, forward propulsors may be acting on a minimum to adjust the direction of the aircraft to the landing location. At net thrust angle **720**, the aircraft has achieved and completed its final stage of descent. In a non-limiting embodiment, the forward propulsor may be turned off and the vertical propulsors may still be operating to slowly land on the landing location. In a non-limiting embodiment, at net thrust angle **720**, in a final stage of descent, the aircraft may be incapable of generating enough lift to counteract gravity. The aircraft may generate an amount of lift that helps slow the descent of the aircraft. Wings of the aircraft may be stalled while also producing a lift factor that slows down the aircraft. The wings may produce attached lift. The aircraft may set down as the attached lift dies out. In the final stage of descent, abortion of the final stage of descent may be impossible. In some embodiments, the aircraft has a thrust to weight ratio of 1 or greater before the final stage of descent. The aircraft may have a thrust to weight ratio of less than 1 but substantially close to 1 (e.g. 0.9) during the final stage of descent.

[0123] Referring now to FIG. **8**, an exemplary embodiment of a machine-learning module **800** that may perform one or more machine-learning processes as described in this disclosure is illustrated. Machine-learning module may perform determinations, classification, and/or analysis steps, methods, processes, or the like as described in this disclosure using machine learning processes. A “machine learning process,” as used in this disclosure, is a process that automatically uses training data **804** to generate an algorithm that will be performed by a computing device/module to produce outputs **808** given data provided as inputs **812**; this is in contrast to a non-machine learning software program where the commands to be executed are determined in advance by a user and written in a programming language.

[0124] Still referring to FIG. **8**, “training data,” as used herein, is data containing correlations that a machine-learning process may use to model relationships between two or more categories of data elements. For instance, and without limitation, training data **804** may include a plurality of data entries, each entry representing a set of data elements that were recorded, received, and/or generated together; data elements may be correlated by shared existence in a given data entry, by proximity in a given data entry, or the like. Multiple data entries in training data **804** may evince one or more trends in correlations between categories of data elements; for instance, and without limitation, a higher value of a first data element belonging to a first category of data element may tend to correlate to a higher value of a second data element belonging to a second category of data element, indicating a possible proportional or other mathematical relationship linking values belonging to the two categories. Multiple categories of data elements may be related in training data **804** according to various correlations; correlations may indicate causative and/or predictive links between categories of data elements, which may be modeled as relationships such as mathematical relationships by machine-learning processes as described in further detail below. Training data **804** may be formatted and/or organized by categories of data elements, for instance by associating data elements with one or more descriptors corresponding to categories of data

elements. As a non-limiting example, training data **804** may include data entered in standardized forms by persons or processes, such that entry of a given data element in a given field in a form may be mapped to one or more descriptors of categories. Elements in training data **804** may be linked to descriptors of categories by tags, tokens, or other data elements; for instance, and without limitation, training data **804** may be provided in fixed-length formats, formats linking positions of data to categories such as comma-separated value (CSV) formats and/or self-describing formats such as extensible markup language (XML), JavaScript Object Notation (JSON), or the like, enabling processes or devices to detect categories of data.

[0125] Alternatively or additionally, and continuing to refer to FIG. **8**, training data **804** may include one or more elements that are not categorized; that is, training data **804** may not be formatted or contain descriptors for some elements of data. Machine-learning algorithms and/or other processes may sort training data **804** according to one or more categorizations using, for instance, natural language processing algorithms, tokenization, detection of correlated values in raw data and the like; categories may be generated using correlation and/or other processing algorithms. As a non-limiting example, in a corpus of text, phrases making up a number “n” of compound words, such as nouns modified by other nouns, may be identified according to a statistically significant prevalence of n-grams containing such words in a particular order; such an n-gram may be categorized as an element of language such as a “word” to be tracked similarly to single words, generating a new category as a result of statistical analysis. Similarly, in a data entry including some textual data, a person's name may be identified by reference to a list, dictionary, or other compendium of terms, permitting ad-hoc categorization by machine-learning algorithms, and/or automated association of data in the data entry with descriptors or into a given format. The ability to categorize data entries automatically may enable the same training data **804** to be made applicable for two or more distinct machine-learning algorithms as described in further detail below. Training data **804** used by machine-learning module **800** may correlate any input data as described in this disclosure to any output data as described in this disclosure. As a non-limiting illustrative example flight elements and/or pilot signals may be inputs, wherein an output may be an autonomous function.

[0126] Further referring to FIG. **8**, training data may be filtered, sorted, and/or selected using one or more supervised and/or unsupervised machine-learning processes and/or models as described in further detail below; such models may include without limitation a training data classifier **816**. Training data classifier **816** may include a “classifier,” which as used in this disclosure is a machine-learning model as defined below, such as a mathematical model, neural net, or program generated by a machine learning algorithm known as a “classification algorithm,” as described in further detail below, that sorts inputs into categories or bins of data, outputting the categories or bins of data and/or labels associated therewith. A classifier may be configured to output at least a datum that labels or otherwise identifies a set of data that are clustered together, found to be close under a distance metric as described below, or the like. Machine-learning module **800** may generate a classifier using a classification algorithm, defined as a process whereby a computing device and/or any module and/or component operating thereon derives a classifier from training data **804**. Classification may be performed using, without limitation, linear classifiers such as without limitation logistic regression and/or naive Bayes classifiers, nearest neighbor classifiers such as k-nearest neighbors classifiers, support vector machines, least squares support vector machines, fisher's linear discriminant, quadratic classifiers, decision trees, boosted trees, random forest classifiers, learning vector quantization, and/or neural network-based classifiers. As a non-limiting example, training data classifier **1616** may classify elements of training data to sub-categories of flight elements such as torques, forces, thrusts, directions, and the like thereof.

[0127] Still referring to FIG. **8**, machine-learning module **800** may be configured to perform a lazy-learning process **820** and/or protocol, which may alternatively be referred to as a “lazy loading” or “call-when-needed” process and/or protocol, may be a process whereby machine learning is

conducted upon receipt of an input to be converted to an output, by combining the input and training set to derive the algorithm to be used to produce the output on demand. For instance, an initial set of simulations may be performed to cover an initial heuristic and/or “first guess” at an output and/or relationship. As a non-limiting example, an initial heuristic may include a ranking of associations between inputs and elements of training data **804**. Heuristic may include selecting some number of highest-ranking associations and/or training data **804** elements. Lazy learning may implement any suitable lazy learning algorithm, including without limitation a K-nearest neighbors algorithm, a lazy naïve Bayes algorithm, or the like; persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various lazy-learning algorithms that may be applied to generate outputs as described in this disclosure, including without limitation lazy learning applications of machine-learning algorithms as described in further detail below.

[0128] Still referring to FIG. **8**, additionally or alternatively, machine-learning processes as described in this disclosure may be used to generate machine-learning models **824**. A “machine-learning model,” as used in this disclosure, is a mathematical and/or algorithmic representation of a relationship between inputs and outputs, as generated using any machine-learning process including without limitation any process as described above and stored in memory; an input is submitted to a machine-learning model **824** once created, which generates an output based on the relationship that was derived. For instance, and without limitation, a linear regression model, generated using a linear regression algorithm, may compute a linear combination of input data using coefficients derived during machine-learning processes to calculate an output datum. As a further non-limiting example, a machine-learning model **824** may be generated by creating an artificial neural network, such as a convolutional neural network comprising an input layer of nodes, one or more intermediate layers, and an output layer of nodes. Connections between nodes may be created via the process of “training” the network, in which elements from a training data **804** set are applied to the input nodes, a suitable training algorithm (such as Levenberg-Marquardt, conjugate gradient, simulated annealing, or other algorithms) is then used to adjust the connections and weights between nodes in adjacent layers of the neural network to produce the desired values at the output nodes. This process is sometimes referred to as deep learning.

[0129] Still referring to FIG. **8**, machine-learning algorithms may include at least a supervised machine-learning process **828**. At least a supervised machine-learning process **828**, as defined herein, include algorithms that receive a training set relating a number of inputs to a number of outputs, and seek to find one or more mathematical relations relating inputs to outputs, where each of the one or more mathematical relations is optimal according to some criterion specified to the algorithm using some scoring function. For instance, a supervised learning algorithm may include flight elements and/or pilot signals as described above as inputs, autonomous functions as outputs, and a scoring function representing a desired form of relationship to be detected between inputs and outputs; scoring function may, for instance, seek to maximize the probability that a given input and/or combination of elements inputs is associated with a given output to minimize the probability that a given input is not associated with a given output. Scoring function may be expressed as a risk function representing an “expected loss” of an algorithm relating inputs to outputs, where loss is computed as an error function representing a degree to which a prediction generated by the relation is incorrect when compared to a given input-output pair provided in training data **804**. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various possible variations of at least a supervised machine-learning process **828** that may be used to determine relation between inputs and outputs. Supervised machine-learning processes may include classification algorithms as defined above.

[0130] Further referring to FIG. **8**, machine learning processes may include at least an unsupervised machine-learning processes **832**. An unsupervised machine-learning process, as used herein, is a process that derives inferences in datasets without regard to labels; as a result, an unsupervised machine-learning process may be free to discover any structure, relationship, and/or correlation

provided in the data. Unsupervised processes may not require a response variable; unsupervised processes may be used to find interesting patterns and/or inferences between variables, to determine a degree of correlation between two or more variables, or the like.

[0131] Still referring to FIG. 8, machine-learning module **800** may be designed and configured to create a machine-learning model **824** using techniques for development of linear regression models. Linear regression models may include ordinary least squares regression, which aims to minimize the square of the difference between predicted outcomes and actual outcomes according to an appropriate norm for measuring such a difference (e.g. a vector-space distance norm); coefficients of the resulting linear equation may be modified to improve minimization. Linear regression models may include ridge regression methods, where the function to be minimized includes the least-squares function plus term multiplying the square of each coefficient by a scalar amount to penalize large coefficients. Linear regression models may include least absolute shrinkage and selection operator (LASSO) models, in which ridge regression is combined with multiplying the least-squares term by a factor of 1 divided by double the number of samples. Linear regression models may include a multi-task lasso model wherein the norm applied in the least-squares term of the lasso model is the Frobenius norm amounting to the square root of the sum of squares of all terms. Linear regression models may include the elastic net model, a multi-task elastic net model, a least angle regression model, a LARS lasso model, an orthogonal matching pursuit model, a Bayesian regression model, a logistic regression model, a stochastic gradient descent model, a perceptron model, a passive aggressive algorithm, a robustness regression model, a Huber regression model, or any other suitable model that may occur to persons skilled in the art upon reviewing the entirety of this disclosure. Linear regression models may be generalized in an embodiment to polynomial regression models, whereby a polynomial equation (e.g. a quadratic, cubic or higher-order equation) providing a best predicted output/actual output fit is sought; similar methods to those described above may be applied to minimize error functions, as will be apparent to persons skilled in the art upon reviewing the entirety of this disclosure.

[0132] Continuing to refer to FIG. 8, machine-learning algorithms may include, without limitation, linear discriminant analysis. Machine-learning algorithm may include quadratic discriminate analysis. Machine-learning algorithms may include kernel ridge regression. Machine-learning algorithms may include support vector machines, including without limitation support vector classification-based regression processes. Machine-learning algorithms may include stochastic gradient descent algorithms, including classification and regression algorithms based on stochastic gradient descent. Machine-learning algorithms may include nearest neighbors algorithms. Machine-learning algorithms may include Gaussian processes such as Gaussian Process Regression. Machine-learning algorithms may include cross-decomposition algorithms, including partial least squares and/or canonical correlation analysis. Machine-learning algorithms may include naïve Bayes methods. Machine-learning algorithms may include algorithms based on decision trees, such as decision tree classification or regression algorithms. Machine-learning algorithms may include ensemble methods such as bagging meta-estimator, forest of randomized trees, AdaBoost, gradient tree boosting, and/or voting classifier methods. Machine-learning algorithms may include neural net algorithms, including convolutional neural net processes.

[0133] It is to be noted that any one or more of the aspects and embodiments described herein may be conveniently implemented using one or more machines (e.g., one or more computing devices that are utilized as a user computing device for an electronic document, one or more server devices, such as a document server, etc.) programmed according to the teachings of the present specification, as will be apparent to those of ordinary skill in the computer art. Appropriate software coding can readily be prepared by skilled programmers based on the teachings of the present disclosure, as will be apparent to those of ordinary skill in the software art. Aspects and implementations discussed above employing software and/or software modules may also include appropriate hardware for assisting in the implementation of the machine executable instructions of

the software and/or software module.

[0134] Such software may be a computer program product that employs a machine-readable storage medium. A machine-readable storage medium may be any medium that is capable of storing and/or encoding a sequence of instructions for execution by a machine (e.g., a computing device) and that causes the machine to perform any one of the methodologies and/or embodiments described herein. Examples of a machine-readable storage medium include, but are not limited to, a magnetic disk, an optical disc (e.g., CD, CD-R, DVD, DVD-R, etc.), a magneto-optical disk, a read-only memory “ROM” device, a random access memory “RAM” device, a magnetic card, an optical card, a solid-state memory device, an EPROM, an EEPROM, and any combinations thereof. A machine-readable medium, as used herein, is intended to include a single medium as well as a collection of physically separate media, such as, for example, a collection of compact discs or one or more hard disk drives in combination with a computer memory. As used herein, a machine-readable storage medium does not include transitory forms of signal transmission.

[0135] Such software may also include information (e.g., data) carried as a data signal on a data carrier, such as a carrier wave. For example, machine-executable information may be included as a data-carrying signal embodied in a data carrier in which the signal encodes a sequence of instruction, or portion thereof, for execution by a machine (e.g., a computing device) and any related information (e.g., data structures and data) that causes the machine to perform any one of the methodologies and/or embodiments described herein.

[0136] Examples of a computing device include, but are not limited to, an electronic book reading device, a computer workstation, a terminal computer, a server computer, a handheld device (e.g., a tablet computer, a smartphone, etc.), a web appliance, a network router, a network switch, a network bridge, any machine capable of executing a sequence of instructions that specify an action to be taken by that machine, and any combinations thereof. In one example, a computing device may include and/or be included in a kiosk.

[0137] FIG. 9 shows a diagrammatic representation of one embodiment of a computing device in the exemplary form of a computer system **900** within which a set of instructions for causing a control system to perform any one or more of the aspects and/or methodologies of the present disclosure may be executed. It is also contemplated that multiple computing devices may be utilized to implement a specially configured set of instructions for causing one or more of the devices to perform any one or more of the aspects and/or methodologies of the present disclosure. Computer system **900** includes a processor **904** and a memory **908** that communicate with each other, and with other components, via a bus **912**. Bus **912** may include any of several types of bus structures including, but not limited to, a memory bus, a memory controller, a peripheral bus, a local bus, and any combinations thereof, using any of a variety of bus architectures.

[0138] Processor **904** may include any suitable processor, such as without limitation a processor incorporating logical circuitry for performing arithmetic and logical operations, such as an arithmetic and logic unit (ALU), which may be regulated with a state machine and directed by operational inputs from memory and/or sensors; processor **904** may be organized according to Von Neumann and/or Harvard architecture as a non-limiting example. Processor **904** may include, incorporate, and/or be incorporated in, without limitation, a microcontroller, microprocessor, digital signal processor (DSP), Field Programmable Gate Array (FPGA), Complex Programmable Logic Device (CPLD), Graphical Processing Unit (GPU), general purpose GPU, Tensor Processing Unit (TPU), analog or mixed signal processor, Trusted Platform Module (TPM), a floating point unit (FPU), and/or system on a chip (SoC)

[0139] Memory **908** may include various components (e.g., machine-readable media) including, but not limited to, a random-access memory component, a read only component, and any combinations thereof. In one example, a basic input/output system **916** (BIOS), including basic routines that help to transfer information between elements within computer system **900**, such as during start-up, may be stored in memory **908**. Memory **908** may also include (e.g., stored on one

or more machine-readable media) instructions (e.g., software) **920** embodying any one or more of the aspects and/or methodologies of the present disclosure. In another example, memory **908** may further include any number of program modules including, but not limited to, an operating system, one or more application programs, other program modules, program data, and any combinations thereof.

[0140] Computer system **900** may also include a storage device **924**. Examples of a storage device (e.g., storage device **924**) include, but are not limited to, a hard disk drive, a magnetic disk drive, an optical disc drive in combination with an optical medium, a solid-state memory device, and any combinations thereof. Storage device **924** may be connected to bus **912** by an appropriate interface (not shown). Example interfaces include, but are not limited to, SCSI, advanced technology attachment (ATA), serial ATA, universal serial bus (USB), IEEE 1394 (FIREWIRE), and any combinations thereof. In one example, storage device **924** (or one or more components thereof) may be removably interfaced with computer system **900** (e.g., via an external port connector (not shown)). Particularly, storage device **924** and an associated machine-readable medium **928** may provide nonvolatile and/or volatile storage of machine-readable instructions, data structures, program modules, and/or other data for computer system **900**. In one example, software **920** may reside, completely or partially, within machine-readable medium **928**. In another example, software **920** may reside, completely or partially, within processor **904**.

[0141] Computer system **900** may also include an input device **932**. In one example, a user of computer system **900** may enter commands and/or other information into computer system **900** via input device **932**. Examples of an input device **932** include, but are not limited to, an alpha-numeric input device (e.g., a keyboard), a pointing device, a joystick, a gamepad, an audio input device (e.g., a microphone, a voice response system, etc.), a cursor control device (e.g., a mouse), a touchpad, an optical scanner, a video capture device (e.g., a still camera, a video camera), a touchscreen, and any combinations thereof. Input device **932** may be interfaced to bus **912** via any of a variety of interfaces (not shown) including, but not limited to, a serial interface, a parallel interface, a game port, a USB interface, a FIREWIRE interface, a direct interface to bus **912**, and any combinations thereof. Input device **932** may include a touch screen interface that may be a part of or separate from display **936**, discussed further below. Input device **932** may be utilized as a user selection device for selecting one or more graphical representations in a graphical interface as described above.

[0142] A user may also input commands and/or other information to computer system **900** via storage device **924** (e.g., a removable disk drive, a flash drive, etc.) and/or network interface device **940**. A network interface device, such as network interface device **940**, may be utilized for connecting computer system **900** to one or more of a variety of networks, such as network **944**, and one or more remote devices **948** connected thereto. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A network, such as network **944**, may employ a wired and/or a wireless mode of communication. In general, any network topology may be used. Information (e.g., data, software **920**, etc.) may be communicated to and/or from computer system **900** via network interface device **940**.

[0143] Computer system **900** may further include a video display adapter **952** for communicating a displayable image to a display device, such as display device **936**. Examples of a display device include, but are not limited to, a liquid crystal display (LCD), a cathode ray tube (CRT), a plasma display, a light emitting diode (LED) display, and any combinations thereof. Display adapter **952**

and display device **936** may be utilized in combination with processor **904** to provide graphical representations of aspects of the present disclosure. In addition to a display device, computer system **900** may include one or more other peripheral output devices including, but not limited to, an audio speaker, a printer, and any combinations thereof. Such peripheral output devices may be connected to bus **912** via a peripheral interface **956**. Examples of a peripheral interface include, but are not limited to, a serial port, a USB connection, a FIREWIRE connection, a parallel connection, and any combinations thereof.

[0144] The foregoing has been a detailed description of illustrative embodiments of the invention. Various modifications and additions can be made without departing from the spirit and scope of this invention. Features of each of the various embodiments described above may be combined with features of other described embodiments as appropriate in order to provide a multiplicity of feature combinations in associated new embodiments. Furthermore, while the foregoing describes a number of separate embodiments, what has been described herein is merely illustrative of the application of the principles of the present invention. Additionally, although particular methods herein may be illustrated and/or described as being performed in a specific order, the ordering is highly variable within ordinary skill to achieve methods, systems, and software according to the present disclosure. Accordingly, this description is meant to be taken only by way of example, and not to otherwise limit the scope of this invention.

[0145] Exemplary embodiments have been disclosed above and illustrated in the accompanying drawings. It will be understood by those skilled in the art that various changes, omissions, and additions may be made to that which is specifically disclosed herein without departing from the spirit and scope of the present invention.

[0146] In the descriptions above and in the claims, phrases such as “at least one of” or “one or more of” may occur followed by a conjunctive list of elements or features. The term “and/or” may also occur in a list of two or more elements or features. Unless otherwise implicitly or explicitly contradicted by the context in which it is used, such a phrase is intended to mean any of the listed elements or features individually or any of the recited elements or features in combination with any of the other recited elements or features. For example, the phrases “at least one of A and B;” “one or more of A and B;” and “A and/or B” are each intended to mean “A alone, B alone, or A and B together.” A similar interpretation is also intended for lists including three or more items. For example, the phrases “at least one of A, B, and C;” “one or more of A, B, and C;” and “A, B, and/or C” are each intended to mean “A alone, B alone, C alone, A and B together, A and C together, B and C together, or A and B and C together.” In addition, use of the term “based on,” above and in the claims is intended to mean, “based at least in part on,” such that an unrecited feature or element is also permissible.

[0147] The subject matter described herein can be embodied in systems, apparatus, methods, and/or articles depending on the desired configuration. The implementations set forth in the foregoing description do not represent all implementations consistent with the subject matter described herein. Instead, they are merely some examples consistent with aspects related to the described subject matter. Although a few variations have been described in detail above, other modifications or additions are possible. In particular, further features and/or variations can be provided in addition to those set forth herein. For example, the implementations described above can be directed to various combinations and sub-combinations of the disclosed features and/or combinations and sub-combinations of several further features disclosed above. In addition, the logic flows depicted in the accompanying figures and/or described herein do not necessarily require the particular order shown, or sequential order, to achieve desirable results. Other implementations may be within the scope of the following claims.

Claims

1. An electric aircraft, wherein the electric aircraft comprises: a fuselage having a fore end, an aft end, a first side, and a second side opposite the first side, and a longitudinal axis running from the fore end to the aft end and between the first side and the second side; a plurality of fixed wings, wherein the plurality of fixed wings includes: a first wing, the first wing including: a first proximal end affixed to the first side of the fuselage; a first distal end projecting away from the fuselage; a middle portion connecting the first proximal end to the first distal end; a first aileron located on the middle portion; and a second wing, the second wing including: a second proximal end affixed to the second side of the fuselage; a second distal end projecting away from the fuselage; a second middle portion connecting the second proximal end to the second distal end; and a second aileron located on the second middle portion; a tail affixed to the aft end of the fuselage; a first boom including a first rear end affixed to the tail and a first forward end proximal to the fore end of the fuselage, wherein the first boom is affixed to the first wing; a second boom including a second rear end affixed to the tail and a second forward end proximal to the fore end of the fuselage, wherein the second boom is affixed to the second wing; and a plurality of lift components, wherein the plurality of lift components includes: at least a first lift component mounted to the first boom; and at least a second lift component mounted to the second boom.
2. The electric aircraft of claim 1, wherein the tail includes: a first vertically projecting element positioned to the first side of the fuselage; and a second vertically projecting element positioned to the second side of the fuselage.
3. The electric aircraft of claim 2, wherein: the first vertically projecting element further comprises at least a first rudder; and the second vertically projecting element further comprises at least a second rudder.
4. The electric aircraft of claim 3, wherein: the first rudder is comprised of a plurality of rudders; and the second rudder is comprised of a plurality of rudders.
5. The electric aircraft of claim 2, wherein: the first rear end is affixed to the first vertically projecting element; and the second rear end is affixed to the second vertically projecting element.
6. The electric aircraft of claim 1, wherein an electric aircraft includes a flight controller configured to operate the plurality of lift components.
7. The electric aircraft of claim 1, wherein each lift component of the plurality of lift components includes a propulsor.
8. The electric aircraft of claim 1, wherein: the first boom is configured to offset torque generated by the at least a first lift component; and the second boom is configured to offset torque generated by the at least a second lift component.
9. The electric aircraft of claim 1, wherein: the first boom is configured to offset a moment generated by the at least a first lift component; and the second boom is configured to offset a moment generated by the at least a second lift component.
10. The electric aircraft of claim 1, wherein the plurality of lift components have a rotational axis offset from a vertical axis by a torque cancellation angle.
11. The electric aircraft of claim 1, wherein each of the lift components is attached to the aircraft at a fixed angle.
12. (canceled)
13. (canceled)
14. The electric aircraft of claim 1, wherein lift components have motors recessed in the first boom or the second boom.
15. (canceled)
16. (canceled)
17. (canceled)
18. (canceled)
19. The electric aircraft of claim 1, further comprising a flight controller configured to turn off lift

propulsors and turn on thrust propulsor for fixed-wing flight.

20. The electric aircraft of claim 1, further comprising a flight controller configured to select between a hover landing and a gliding landing.

21. An electric aircraft, wherein the electric aircraft is comprised of: a plurality of flight components wherein the plurality of flight components comprises: a plurality of control surfaces; a plurality of lift propulsors; at least a thrust propulsor; and a plurality of electric motors configured to power the plurality of lift propulsors and the at least a thrust propulsor; and a flight controller, wherein flight controller is: communicatively connected to a pilot input and flight components; and configured to: receive control datum from a pilot input; and generate an output datum as a function of the control datum.

22. The electric aircraft of claim 21, wherein the plurality of flight components are attached to a plurality of fixed wings.

23. The electric aircraft of claim 22, wherein there are at least two control surfaces mechanically attached to each fixed wing.

24. (canceled)

25. (canceled)

26. (canceled)

27. (canceled)

28. (canceled)

29. (canceled)

30. The electric aircraft of claim 21, wherein a machine learning process is used to generate control surface datum as a function of pilot input training data.

31. The electric aircraft of claim 21, wherein control surface datum is an electrical signal configured to control the plurality of flight components.

32. The electric aircraft of claim 21, wherein the plurality of lift propulsors and the at least a thrust propulsor are separate flight components.

33. (canceled)

34. (canceled)

35. (canceled)

36. (canceled)

37. (canceled)

38. (canceled)

39. (canceled)

40. (canceled)

41. (canceled)
